

# Generalized interpolation-supplemented cascaded lattice Boltzmann method for noise radiated from a circular cylinder

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## ABSTRACT

Flow past a stationary or vibrating body usually leads to intense noise radiation. In order to study this problem via direct simulations, a cascaded lattice Boltzmann method (CLBM) is developed in the moving frame of reference, and implemented in a body-fitted grid using the generalized interpolation-supplemented particle streaming. A far-field perfectly matched layer is also incorporated so as to avoid sound reflection. The proposed generalized interpolation-supplemented cascaded lattice Boltzmann method (GICLBM) is then validated through numerical experiments concerning fixed and forced oscillating circular cylinders, in terms of forces exerting on the cylinder, wall stress, near-field flow dynamics, and far-field sound radiation. Very good consistency is observed for all quantities discussed therein with previous benchmark tests. Furthermore, the noise radiated from a circular cylinder undergoing vortex-induced vibration (VIV) is investigated using the GICLBM, providing interesting results. Firstly, at a mass ratio of  $m^* = 2$ , the flow and vibration dynamics are found to be dependent on the Mach number, and the critical value is found to be approximately  $Ma = 0.1$ . Secondly, through comparisons with scenarios involving a fixed cylinder and a forced vibrating cylinder, it is noted that the VIV generates significantly more complicated sound sources including monopole and drag dipole. Thirdly, in the lock-in condition, the increase in the reduced velocity only alters the magnitude of the radiated sound pressure but not the directivity. Through this study, a high-fidelity, efficient, and relatively simple framework for fluid-structure-sound interactions is proposed.

## 1. Introduction

Flow-induced noise is becoming more and more important in engineering applications, such as aircraft landing gear, high-speed rail, and underwater vehicles. The generation and propagation of noise are closely related to unsteady flows [54]. Especially when the structure is undergoing vibration, interactions between fluid and structure radiate intense noise. Therefore, it is important to investigate flow-induced vibration (FIV) and its noise generation and propagation effects.

Numerical simulations play an important role in acoustic problems. The solutions in computational aeroacoustics (CAA) can generally be divided into direct and hybrid methods [54]. The direct method is to solve the equations of compressible flow in one

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step. In the hybrid method, the sound generation and propagation are decoupled, and the far-field noise is obtained using the flow field information [49]. As the direct method accounts for the interaction between the flow and noise, it is a more suitable way to investigate noise mechanism. In order to investigate the FIV and its associated noise radiation, the direct method needs to include the following factors [41][51]: (1) a low-dispersion and low-dissipation discrete scheme to minimize the error in sound propagation, (2) an appropriate grid to reduce the amount of nodes, (3) a suitable moving boundary condition to alleviate spurious noise, and (4) a non-reflective boundary condition to avoid noise reflection from the boundary.

The lattice Boltzmann method (LBM) on a uniform grid has been proven to be a low-dissipation and low-dispersion method that is suitable for direct acoustic calculations. Therefore, it has been considered a promising method for CAA [41]. However, there is usually a scale difference between acoustic and fluid dynamics simulations. In nonlinear sound source regions that involve complicated fluid interactions, fine grids are needed to resolve the small-scale flow structure. Meanwhile, in regions in which sound waves linearly propagate, only relatively coarse grids are needed to ensure the sound wave does not undergo dispersion and dissipation. Hence, using a uniform grid in both linear propagation regions and nonlinear sound source regions would be wasteful. Moreover, the area of interest in CAA simulations is usually far from the sound source, and the computational domains in CAA are usually much larger than those in computational fluid dynamics. Hence, it takes a larger amount of computing resources to obtain noise on a uniform grid via the direct method, which becomes almost unacceptable for a high Reynolds number ( $Re$ ). A common strategy in the LBM is to refine Cartesian grids and to couple fine and coarse grids by spatial interpolation. The coupling algorithm was implemented on cell vertices by Filippova and Hänel [14] and on cell centers by Rohde et al. [47] and Chen et al. [9]. The spurious noise of the coupling algorithm can be evaluated by calculating a vortex crossing the interface [17]. Astoul et al. [2] developed an “aeroacoustically-compliant” algorithm that is implemented by directly coupling the grids without setting the overlap area. However, this increasingly delicate algorithm comes at a prohibitive cost due to high-order interpolation. More recently, Bellotti et al. [4,5] developed an adaptive multi-resolution approach for LBM and improved the computational efficiency by performing the collision only on “complete leaves” of the tree data structure. The precision can be ensured by the error control with wavelet transform. And the performance in solving hyperbolic equations makes this approach a potential solution to the sound propagation problem. Another problem is that improper handling of the curved boundary may cause deviations in the acoustic scattering calculation. Several methods of curved boundary processing have been proposed, such as bounce back schemes [39], interpolation [57], and extrapolation [18] of the velocity at boundary nodes. Moreover, the curved no-slip boundary condition [52], which is a generalization of the straight solid boundary condition [35], was utilized in the simulation of noise generated by a circular cylinder by Astoul et al. [2]. However, for complex geometry, a large number of grid cells are needed to reconstruct the curved boundary. Another widely used method is the immersed boundary (IB) method, which was proposed by Peskin [43]. Sun et al. [50] tested several benchmark problems to validate the performance of the IB method in CAA simulations. In the IB method, the extra consumption due to interpolation from adjacent nodes is non-negligible. Applying the LBM to a non-uniform and body-fitted grid is a feasible solution because such a grid can commendably balance the requirement in resolving near-field flow dynamics and far-field sound propagation, and can eliminate the spurious noise caused by the improper curved boundary treatment.

The interpolation-supplemented lattice Boltzmann method (ISLBM), which was proposed by He et al. [21], has been successfully implemented on a non-uniform grid with a curved boundary [20]. The classical LBM includes streaming and collision processes. The nonlinear convective term in the Navier–Stokes (NS) equation is computed locally through a collision operation, which significantly improves the parallelism of simulations. In the ISLBM, particles collide locally and stream on grids with different sizes depending on the reconstruction of the particle streaming velocity. The global time step is restricted by the Courant–Friedrichs–Lewy (CFL) condition so as to satisfy the numerical stability criterion. Based on the ISLBM, Imamura et al. [23] extended this method to arbitrarily structured grids as a generalized formulation of the interpolation-supplemented lattice Boltzmann method (GILBM). The GILBM has been successfully implemented in flows past a bluff body [24,37], non-Newtonian fluids [12], and heat transfer problems [60,61]. The GILBM allows for a flexible configuration of the grid depending on the characteristics of noise generation and propagation and is, therefore, a promising strategy for CAA. Moreover, the GILBM was developed based on the Bhatnagar–Gross–Krook (BGK) model [6], which is a classical collision model with a single relaxation time (SRT). However, for problems with high  $Re$  and  $Ma$ , the numerical stability of the BGK model needs to be improved. A multi-relaxation-time (MRT) method was proposed by d’Humières [11] that has a collision process executed in moment space. A set of orthogonal vectors is constructed, and the equilibrium distribution functions are transformed into moment space. One significant difference between SRT and MRT models is that raw moments correspond to different relaxation times, which decouples the bulk and shear viscosities and improves numerical stability. Li et al. [37] embedded the MRT model in the GILBM and calculated the flow past an airfoil and flat plate problems at high  $Re$ , but other collision models have not been implemented. For example, by constructing higher-order moments from lower-order and macroscopic variables, the cascaded LBM (CLBM), which was proposed by Geier et al. [15], executes the collision process in central moment space and maintains Galilean invariance. Furthermore, there are many other efficient collision models such as the entropic model [1] and the regularized model [34]. Overall, the GILBM is a promising approach for CAA, but its accuracy has not yet been verified for acoustic problems.

Modeling sound generation from a moving boundary is very important for the problem of FIV and its associated noise radiation. The curved boundary method, which is mainly applied to stationary problems, can produce a non-negligible spurious pressure oscillation that can lead to false acoustic results [33]. Therefore, Lallemand and Luo [33] proposed an overset method based on interpolation in moment space for a moving boundary. However, the spurious pressure oscillations in the far-field generated by this method have not yet been assessed. On a non-uniform grid, mesh reconstruction is another efficient approach for a moving boundary and is usually employed. However, in order to avoid tedious interpolation, the interpolated stencil in the GILBM is calculated during initialization, and the subsequent calculation is read directly from the memory. Hence, the mesh reconstruction process, which needs to update the positions of nodes and interpolation stencils, is very time-consuming. The moving frame of reference algorithm

proposed by Li et al. [38] is an effective method for modeling the moving rigid structure without mesh reconstruction. Using a coordinate transformation, the Navier–Stokes (NS) equations in the absolute frame of reference are transformed into the moving frame of reference. Since no other hypothesis is introduced in the derivation, it can be deduced that the far-field noise will not be polluted.

In addition, the computation of noise radiation is usually carried out in the open domain. In contrast, the LBM is implemented in the finite domain. A sound wave reflected from the boundary will pollute the inner sound region, so appropriate boundary conditions are required to avoid reflective noise. There are two common solutions. The first is that the sound waves are allowed to pass through the boundary without reflection, in which case the typical method is a non-reflecting characteristics-based boundary condition (NSCBC) [28]. The other is adding a non-reflective sound boundary layer where sound waves from all directions are dissipated, and the typical methods for this include an absorbing boundary condition (ABC) [56,10] or the perfectly matched layer (PML) technique [42]. Kam et al. [28] evaluated the performance of several non-reflective boundary strategies, including NSCBC and ABC, and found that ABC worked better. It should be noted that the non-reflective sound boundary layer is implemented with an extended computational domain, so more grid cells are needed. Najafi-Yazdi and Mongeau [42] integrated the PML method in LBM with PML formulations obtained based on the lattice Boltzmann equations. In the PML, sound waves from all directions can undergo exponential dissipation. In fact, the damping term in ABC is included in PML formulations. Therefore, it is reasonable to expect that the PML method performs better.

The goal of this paper is to propose a framework for calculating FIV and its associated noise radiation. The CLBM is developed in the moving frame of reference and embedded in the GILBM on a non-uniform grid. Moreover, the perfectly matched layer (PML) [42] technique is adopted to minimize reflected noise from the boundary of the computational domain. The accuracy of the GICLBM for noise radiation is validated for fixed and forced oscillating circular cylinders. A circular cylinder undergoing VIV is also investigated using the present GICLBM.

## 2. Methodology

### 2.1. Generalized interpolation-supplemented lattice Boltzmann method with cascaded collision

The classical LBM, in particular, the two-dimensional scheme with nine discrete velocities (D2Q9) [45] with three conserved moments, is used to approximate the solution of the weakly compressible NS equations. In the LBGK model [6], all the distribution functions are relaxed with a SRT. Beyond the LBGK model, the collision operator can also be carried out with the MRT approach [11], in which the distribution functions are relaxed with different relaxation times in moment space. Lallemand and Luo [32] have proven that the MRT-LBM can significantly improve numerical stability [32]. Furthermore, shifting from raw moments to central moments can transform the MRT-LBM into the CLBM. The generalized formulation of the LBM is carried out in two independent steps:

$$\begin{aligned} \text{collision: } f_i^*(\mathbf{x}, t) &= f_i(\mathbf{x}, t) - \mathcal{B}_{i,j} \left[ f_j(\mathbf{x}, t) - f_j^{eq}(\mathbf{x}, t) \right] + (I_{i,j} - A_{i,j}/2) \hat{R}_j(\mathbf{x}, t) \Delta t, \\ \text{streaming: } f_i(\mathbf{x}, t + \Delta t) &= f_i^*(\mathbf{x} - \mathbf{e}_i \Delta t, t), \end{aligned} \quad (1)$$

where  $f_i(\mathbf{x}, t)$  is the distribution function and  $f_i^*(\mathbf{x}, t)$  represents the post-collision distribution function. Herein the Einstein notation with summation on the repeated indices is utilized. The generalized collision matrix  $\mathcal{B}$  has different forms corresponding to various collision models. The letter  $\mathbf{I}$  is an identity matrix. The letter  $\mathbf{x}$  is the lattice position vector and  $t$  is the time. In general, for the LBM on a uniform grid,  $\Delta \mathbf{x} = 1$  and  $\Delta t = 1$ . For two-dimensional problems, the discrete velocity  $\mathbf{e}$  includes nine velocity components for the D2Q9 scheme, and is expressed as

$$\mathbf{e} = c \begin{bmatrix} 0 & 1 & 0 & -1 & 0 & 1 & -1 & -1 & 1 \\ 0 & 0 & 1 & 0 & -1 & 1 & 1 & -1 & -1 \end{bmatrix}, \quad (2)$$

where the unit of  $\mathbf{e}_i$  is the lattice speed  $c = \Delta x / \Delta t$ . The equilibrium distribution function  $f_i^{eq}$  in the D2Q9 scheme is

$$f_i^{eq} = w_i \rho \left[ 1 + \frac{\mathbf{e}_i \cdot \mathbf{u}}{c_s^2} + \frac{(\mathbf{e}_i \cdot \mathbf{u})^2}{2c_s^4} - \frac{\mathbf{u}^2}{2c_s^2} \right] \quad (3)$$

where the weighting factor is  $w_0 = 4/9$ ,  $w_i = 1/9$  for  $i = 1, 2, 3, 4$  and  $w_i = 1/36$  for  $i = 5, 6, 7, 8$ . The letter  $\hat{R}$  denotes the external forcing term. The macroscopic density  $\rho$  and velocity  $\mathbf{u} = [u_x, u_y]^T$  can be obtained via

$$\rho = \sum_{i=0}^8 f_i, \quad \rho \mathbf{u} = \sum_{i=0}^8 \mathbf{e}_i f_i + \frac{1}{2} \tilde{\mathbf{F}} \Delta t, \quad (4)$$

where  $\tilde{\mathbf{F}} = [\tilde{F}_x, \tilde{F}_y]^T$  is the body force vector. Under the isothermal condition, the pressure is expressed as  $p = (\rho - \rho_0) c_s^2$ , where  $c_s$  is the sound speed defined as  $c_s = c / \sqrt{3}$  [45] and  $\rho_0 = 1$  denotes the reference fluid density.

The CLBM, which executes collisions in the central moment space, can maintain Galilean invariance to improve numerical stability [15,16]. Fei and Luo [13] demonstrated the relationship between the MRT-LBM and the CLBM, and proposed a new derivation for the CLBM. The governing equation of the CLBM model without the forcing term is given as

$$f_i(\mathbf{x} + \mathbf{e}_i \Delta t, t + \Delta t) = \mathbf{M}_{i,j}^{-1} \mathbf{N}_{i,j}^{-1} \left[ (\mathbf{I}_{i,j} - \mathbf{\Lambda}_{i,j}) T_j(\mathbf{x}, t) + \mathbf{\Lambda}_{i,j} T_j^{eq}(\mathbf{x}, t) \right], \quad (5)$$

where  $T_i$  is the combination of distribution functions in central moment space. The discrete distribution function in central moment space is defined as

$$k_{mn} = \sum_{i=0}^8 f_i (e_{ix} - u_x)^m (e_{iy} - u_y)^n. \quad (6)$$

And  $T_i$  is expressed as

$$\mathbf{T} = [k_{00}, k_{10}, k_{01}, k_{20} + k_{02}, k_{20} - k_{02}, k_{11}, k_{21}, k_{12}, k_{22}]^T. \quad (7)$$

This can also be derived in another way proposed by Fei and Luo [13], where  $\mathbf{T} = \mathbf{N} \mathbf{M} \mathbf{f}$  is given. Both  $\mathbf{M}$  and  $\mathbf{N}$  are  $9 \times 9$  transforming matrices for the D2Q9 scheme. A transformation is first carried out to convert from the velocity space to the moment space using  $\mathbf{M}$ , and then to central moment space using the shift matrix  $\mathbf{N}$ . Details of the above transformation matrix  $\mathbf{M}$  and  $\mathbf{N}$  can be found in Ref. [13].

The corresponding  $k_{mn}^{eq}$  can be obtained from Eq. (6). As suggested by Premnath and Banerjee [44],  $T_i^{eq}$  is obtained from the continuous Maxwell–Boltzmann distribution in velocity space  $\mathbf{e} = [e_x, e_y]^T$ , and can be given as

$$T_i^{eq} = \mathbf{N} \mathbf{M} \mathbf{f}^M = \rho [1, 0, 0, 2c_s^2, 0, 0, 0, 0, 2c_s^4]^T, \quad (8)$$

where the continuous Maxwell–Boltzmann distribution is expressed as

$$\mathbf{f}^M = \frac{\rho}{2\pi c_s^2} \exp \left[ -\frac{(\mathbf{e} - \mathbf{u})^2}{2c_s^2} \right]. \quad (9)$$

The relaxation matrix is expressed as  $\mathbf{\Lambda} = \text{diag}(s_0, s_1, s_2, s_3, s_4, s_5, s_6, s_7, s_8)$ , and needs to satisfy  $s_1 = s_2$ ,  $s_4 = s_5$  and  $s_6 = s_7$  in the CLBM. The first and third order central moments will be relaxed to the equilibrium state with the same relaxation parameters, such that we have  $s_1 = s_2$  and  $s_6 = s_7$ . In the meantime, because the zeroth and first order moments are conserved components and they are not updated in the collision step, the corresponding relaxation parameters are set to be 1. The relaxation of the third and fourth order moments plays a significant role in the numerical stability, and being capable of freely adjusting these values is the main advantage of the cascaded collision model. For the second order moment, the bulk viscosity and shear viscosity can be adjusted with different relaxation parameters. In order to ensure an isotropic viscosity, the corresponding relaxation parameters should satisfy  $s_4 = s_5$  [44].

Moreover, the forcing terms  $\hat{R}_i$  in Eq. (1) must also be adopted in central moment space. Premnath and Banerjee [44] and Fei and Luo [13] both proposed forcing term schemes for the CLBM model, and Eq. (5) can be rewritten as

$$f_i(\mathbf{x} + \mathbf{e}_i \Delta t, t + \Delta t) = \mathbf{M}_{i,j}^{-1} \mathbf{N}_{i,j}^{-1} \left[ (\mathbf{I}_{i,j} - \mathbf{\Lambda}_{i,j}) T_j(\mathbf{x}, t) + \mathbf{\Lambda}_{i,j} T_j^{eq}(\mathbf{x}, t) + (\mathbf{I}_{i,j} - \mathbf{\Lambda}_{i,j}/2) \hat{F}_j(\mathbf{x}, t) \Delta t \right], \quad (10)$$

where  $\hat{F}_i$  is derived from  $\hat{\mathbf{F}} = \mathbf{N} \mathbf{M} \hat{\mathbf{R}}$ . As shown by Fei and Luo [13],  $\hat{R}_i$  is defined as

$$\hat{R}_i = \frac{\tilde{\mathbf{F}}}{\rho} \cdot \frac{(\mathbf{e}_i - \mathbf{u})}{c_s^2} \mathbf{f}_i^M. \quad (11)$$

Thus,  $\hat{F}_i$  can be expressed as

$$\hat{F}_i(\mathbf{x}, t) = [0, \tilde{F}_x, \tilde{F}_y, 0, 0, 0, c_s^2 \tilde{F}_y, c_s^2 \tilde{F}_x, 0]^T. \quad (12)$$

As a result, the collision step for the CLBM model is

$$f_i^*(\mathbf{x}, t) = \mathbf{M}_{i,j}^{-1} \mathbf{N}_{i,j}^{-1} \left[ (\mathbf{I}_{i,j} - \mathbf{\Lambda}_{i,j}) T_j(\mathbf{x}, t) + \mathbf{\Lambda}_{i,j} T_j^{eq}(\mathbf{x}, t) + (\mathbf{I}_{i,j} - \mathbf{\Lambda}_{i,j}/2) \hat{F}_j(\mathbf{x}, t) \Delta t \right]. \quad (13)$$

To extend beyond the LBM, which is calculated on a uniform grid, to a body-fitted grid, Imamura et al. [23] proposed the GILBM. As shown in Fig. 1,  $\mathbf{x} = (x, y)$  and  $\xi = (\xi, \eta)$  represent a physical domain with generalized coordinates and a computational domain with orthogonal coordinates, respectively. The coordinate transformation is

$$\begin{bmatrix} \partial_x \xi & \partial_y \xi \\ \partial_x \eta & \partial_y \eta \end{bmatrix} = \begin{bmatrix} \partial_\eta y & -\partial_\eta x \\ -\partial_\xi y & \partial_\xi x \end{bmatrix} \mathbf{J}^{-1}, \quad (14)$$

where the Jacobian  $\mathbf{J}$  is also the area ratio of the same cell in the above two domains and is calculated as

$$\mathbf{J} = \partial_\xi x \partial_\eta y - \partial_\eta x \partial_\xi y. \quad (15)$$

Due to the collision step of each node being carried out independently, Eq. (13) can be transformed from the physical domain to the computational domain by directly replacing  $\mathbf{x}$  with  $\xi$ . Therefore, the collision step in the GILBM can be expressed as

$$f_i^*(\xi, t) = \mathbf{M}^{-1} \mathbf{N}^{-1} \left[ (\mathbf{I} - \mathbf{\Lambda}) T_j(\xi, t) + \mathbf{\Lambda} T_j^{eq}(\xi, t) + (\mathbf{I} - \mathbf{\Lambda}) \hat{F}_j(\xi, t) \Delta t \right]. \quad (16)$$

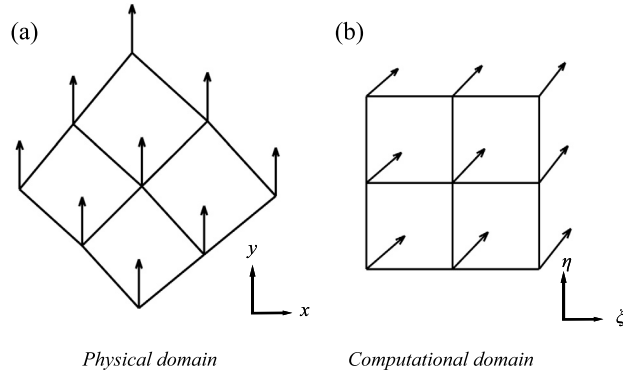


Fig. 1. The discrete velocity (0, 1) in (a) the physical domain and (b) the computational domain.

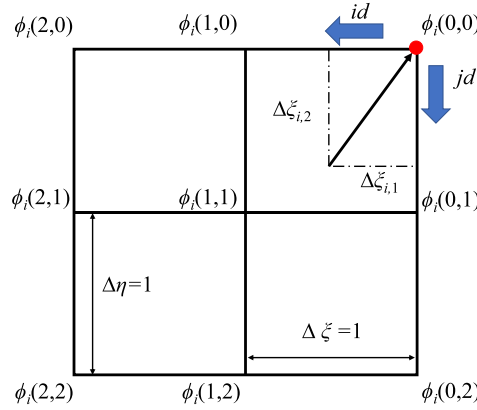


Fig. 2. Stencils utilized for second-order upwind quadratic interpolation.

However, the streaming distance  $e_i \Delta t$  of the distribution function in Eq. (1) is then invalid. In order to ensure that the streaming step is executed synchronously in the physical domain, the contravariant velocity  $\tilde{e}_i$ , which is related to the streaming distance, is utilized in the computational domain. It is defined as

$$(\tilde{e}_{i,1}, \tilde{e}_{i,2}) = (e_{i,1}, e_{i,2}) \begin{bmatrix} \partial_x \xi & \partial_x \eta \\ \partial_y \xi & \partial_y \eta \end{bmatrix}. \quad (17)$$

The streaming step for the GICLBM is modified to

$$f_i(\xi, t + \Delta t) = f_i^*(\xi_i - \Delta \xi_i, t), \quad (18)$$

where  $\Delta \xi_i$  can be obtained by integrating with respect to  $\tilde{e}_i$ . The two-step Runge–Kutta method has been confirmed to be sufficient as an integration scheme [23]. Therefore,  $\Delta \xi_i$  can be calculated using the following two steps,

$$\begin{aligned} \text{step I: } \Delta \xi_i^{(1)} &= \frac{1}{2} \Delta t \tilde{e}_i(\xi), \\ \text{step II: } \Delta \xi_i &= \Delta t \tilde{e}_i(\xi_i - \Delta \xi_i^{(1)}) + O(\Delta t^3). \end{aligned} \quad (19)$$

Consequently, the right-hand side (RHS) of Eq. (18) can be obtained. It should be noted that both  $\tilde{e}_i(\xi - \Delta \xi_i^{(1)})$  and  $f_i^*(\xi - \Delta \xi_i, t)$  are calculated by interpolation with second-order accuracy in space to avoid spurious numerical errors. In particular, second-order upwind quadratic interpolation is utilized for the internal grid, as suggested by He and Doolen [20]. The stencils utilized for interpolation are depicted in Fig. 2. The spatial interpolation of the variables  $\phi$  can be expressed as

$$\phi_i(\xi_i - \Delta \xi_i) = \sum_{k=0}^2 \sum_{l=0}^2 a_{i,k,2} a_{i,l,1} \phi_i(\xi + k \cdot i_d, \eta + l \cdot j_d), \quad (20)$$

where  $i_d$  and  $j_d$  are the upwind direction. The interpolation coefficients are given as

$$\begin{aligned} a_{i,0,\alpha} &= \frac{1}{2} (|\Delta \xi_{i,\alpha}| - 1) (|\Delta \xi_{i,\alpha}| - 2), \\ a_{i,1,\alpha} &= -|\Delta \xi_{i,\alpha}| (|\Delta \xi_{i,\alpha}| - 2), \\ a_{i,2,\alpha} &= \frac{1}{2} |\Delta \xi_{i,\alpha}| (|\Delta \xi_{i,\alpha}| - 1), \end{aligned} \quad (21)$$

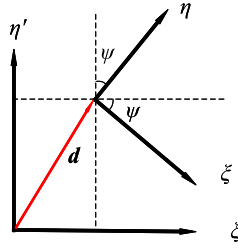


Fig. 3. The coordinate transformation from the absolute frame of reference to the moving frame of reference.

where  $\alpha = 1, 2$  for the D2Q9 scheme and denote  $\xi, \eta$  in the computational domain, respectively. The nodes adjacent to the boundary are calculated as

$$\phi_i(\xi_i - \Delta \xi_i) = \sum_{k=-1}^1 \sum_{l=-1}^1 a_{i,k,2} a_{i,l,1} \phi_i(\xi + k, \eta + l), \quad (22)$$

where the interpolation coefficients are

$$\begin{aligned} a_{i,-1,\alpha} &= \frac{1}{2} \Delta \xi_{i,\alpha} (\Delta \xi_{i,\alpha} - 1), \\ a_{i,0,\alpha} &= 1 - \Delta \xi_{i,\alpha}^2, \\ a_{i,1,\alpha} &= \frac{1}{2} \Delta \xi_{i,\alpha} (\Delta \xi_{i,\alpha} + 1). \end{aligned} \quad (23)$$

Note that the interpolation procedure cannot guarantee local numerical stability because it is merely an affine transformation of distribution functions, while the accuracy of the interpolation can be improved through refining the local grid to avoid sharp spatial variations in the distribution functions. Because the grid size and the contravariant velocities are different on each node, the time step  $\Delta t$  should satisfy the CFL condition to ensure numerical stability [36], i.e.,

$$\Delta t = \lambda \min_{i,a} \left| \frac{1}{\tilde{e}_{i,a}} \right|, \quad (24)$$

where the CFL number is satisfied  $\lambda \in (0, 1]$ . Because  $\Delta t$  is usually not 1 for the GICLBM, the shear viscosity  $\nu$  and the bulk viscosity  $\zeta$  are divided into

$$\nu = c_s^2 \left( \frac{1}{s_4} - \frac{1}{2} \right) \Delta t, \quad \zeta = c_s^2 \left( \frac{1}{s_3} - \frac{1}{2} \right) \Delta t. \quad (25)$$

Thus  $s_4$  is determined from the shear viscosity  $\nu$  and  $\Delta t$ .

## 2.2. Moving frame of reference for a moving rigid structure

Noise radiation from rigid structures, especially flow-induced vibrating structures, is quite common in realistic applications. These FIV problems are handled using the governing equations for fluid dynamics and the motion of structures. In this study, the fluid dynamic forces acting on the structure are first computed by solving Eq. (16), and then the response of the structure to external forces is determined in the next stage. This decoupled method is also called a “loose” coupling algorithm. Since the computational domain is altered as the structure moves, the most general strategy is to reconstruct the grid. However, the calculation of the contravariant velocity and its integration are expensive and usually completed at initialization. Grid reconstruction is usually quite time-consuming, thus the moving frame of the reference algorithm [38] is adopted here to tackle moving boundaries in the GICLBM. The moving frame of reference is substituted for the absolute frame of reference and moves with the rigid structure.

As shown in Fig. 3, the coordinate transformation satisfies

$$\xi' = \mathbf{A} \xi + \mathbf{d}, \quad (26)$$

where  $\xi'$  and  $\xi$  are the coordinates in the absolute frame of reference and moving frame of reference, respectively.  $\mathbf{d}$  denotes the displacement of the rigid structure in the absolute frame of reference. The rotation matrix  $\mathbf{A}$  is formulated as

$$\mathbf{A} = \begin{bmatrix} \cos \psi & \sin \psi \\ -\sin \psi & \cos \psi \end{bmatrix}, \quad (27)$$

where  $\psi$  is the structure's rotational angle. In the most typical case, the weakly compressible NS equation in the absolute reference of frame is expressed as

$$\begin{aligned} \frac{\partial \rho'}{\partial t} + \nabla' \cdot (\rho' \mathbf{u}') &= 0, \\ \frac{\partial \rho' \mathbf{u}'}{\partial t} + \nabla' \cdot (\rho' \mathbf{u}' \mathbf{u}') &= -\nabla' p' + \nabla' \cdot \left[ \rho' \nu \left( \nabla' \mathbf{u}' + (\nabla' \mathbf{u}')^T \right) \right], \end{aligned} \quad (28)$$

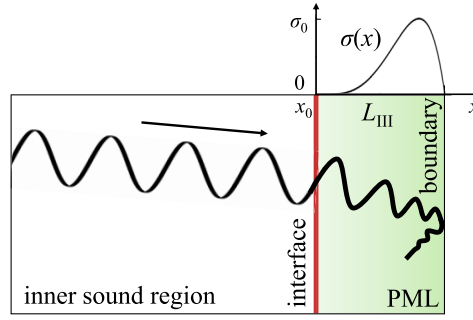


Fig. 4. A schematic diagram of the PML.

where  $\rho'$  and  $\mathbf{u}'$  are flow field variables in the absolute reference of frame. After transformation, the NS equations in the moving frame of reference are modified into [38]

$$\begin{aligned} \frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) &= 0, \\ \frac{\partial \rho \mathbf{u}}{\partial t} + \nabla \cdot (\rho \mathbf{u} \mathbf{u}) &= -\nabla p + \nabla \cdot [\rho \nu (\nabla \mathbf{u} + (\nabla \mathbf{u})^T)] + \rho \mathbf{G}(\mathbf{u}, t), \\ \mathbf{G}(\mathbf{u}, t) &= 2\dot{\psi} \mathbf{I} \mathbf{u} + (\dot{\psi})^2 \xi + \dot{\psi} \mathbf{I} \xi - \mathbf{A}^T \ddot{\mathbf{d}}. \end{aligned} \quad (29)$$

Because the rotational motion is not involved in the present study, the related terms are neglected. Then the rotation matrix  $\mathbf{A}$  becomes an identity matrix and the related terms with  $\dot{\psi}$  and  $\ddot{\psi}$  are removed. Similarly, the forcing term ignoring the rotational motion can be derived from Eq. (12) and expressed as

$$\tilde{\mathbf{F}} = \rho \mathbf{G} \Delta t = -\rho \ddot{\mathbf{d}} \Delta t. \quad (30)$$

The acceleration  $\ddot{\mathbf{d}}$  can be obtained by solving the governing equation of motion for a rigid structure [38],

$$m \ddot{\mathbf{d}} + D \dot{\mathbf{d}} + K \mathbf{d} = \mathbf{F}, \quad (31)$$

where  $m$ ,  $D$ , and  $K$  are the mass, damping, and stiffness, respectively.  $\mathbf{F} = [F_x, F_y]^T$  is the external force exerted on the structure. Moreover, with the lift force exerting on the rigid structure, the displacement and velocity can also be computed via solving the kinematic equation, i.e., Eq. (31).

It should be noted that the boundary conditions also vary with time in the moving reference frame. In the present work, the non-equilibrium extrapolation scheme [19] is used to enforce the no-slip and no-penetration wall boundary and the far-field Dirichlet boundary. Without rotation of the rigid structure, the velocity  $\mathbf{u}_0' = [U_0, 0]^T$  in the far-field boundary condition can be modified to

$$\mathbf{u}_0 = \mathbf{u}_0' - \dot{\mathbf{d}}. \quad (32)$$

### 2.3. Non-reflective far-field buffer layer

The general boundary conditions used for computational fluid dynamics reflect spurious sound waves. Non-reflective boundary conditions are one of the crucial aspects of the CAA. One possible method is similar to the proportional control, which adjusts the sound wave to a reference value before it reaches the boundary of the computational domain [56,10]. Moreover, Najafi-Yazdi and Mongeau [42] proposed a non-reflective boundary condition based on the perfectly matched layer (PML) [22] for the LBM. As an extra term in the collision operator, the PML can theoretically absorb sound waves with any frequency from any direction.

As shown in Fig. 4, an additional non-reflective buffer layer is implemented in the outer zone, which is usually thicker than the wavelength of sound. Within this zone, the collision in Eq. (16) is modified as

$$f_i(\mathbf{x} + \mathbf{e}_i \Delta t, t + \Delta t) = \mathbf{M}^{-1} \mathbf{N}^{-1} \left[ (\mathbf{I} - \Lambda) T_j(\mathbf{x}, t) + \Lambda T_j^{eq}(\mathbf{x}, t) + (\mathbf{I} - \Lambda) \hat{F}_j(\mathbf{x}, t) \Delta t \right] + \Omega_i^{PML}, \quad (33)$$

where

$$\Omega_i^{PML} = -\sigma \left[ \mathbf{e}_i \cdot \nabla \mathbf{Q}_i + 2 \left( f_i^{eq}(\rho, \mathbf{u}) - f_i^{eq}(\rho_0, \mathbf{u}_0) \right) + \sigma \mathbf{Q}_i \right], \quad (34)$$

and the additional term  $\mathbf{Q}$  is calculated as

$$\frac{\partial \mathbf{Q}_i}{\partial t} = f_i^{eq}(\rho, \mathbf{u}) - f_i^{eq}(\rho_0, \mathbf{u}_0). \quad (35)$$

In Eq. (34), the divergence  $\nabla \mathbf{Q}$  is obtained using the second-order central difference scheme and the time marching is carried out using the first-order Euler method. The letter  $\sigma$  is the damping coefficient to be specified. It has been reported that a uniform  $\sigma$  produces some unwanted reflection at the interface of the inner zone and the buffer layer [26].



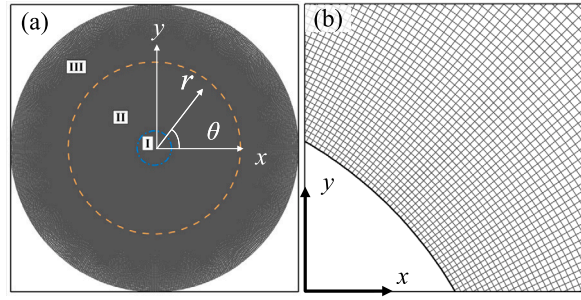


Fig. 5. The body-fitted grid in the physical domain. (a) Regions I, II and III represent different nodes distribution and (b) computational grids which are adjacent to the circular cylinder.

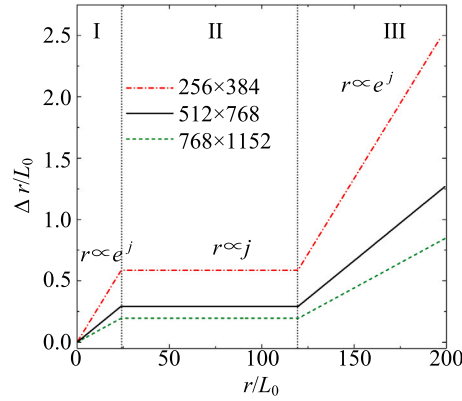


Fig. 6. The node distribution along the radial direction, where  $j$  represents the number of cells.

Due to the similar concepts behind the traditional ABC and the PML, the profile for  $\sigma$  recommended by Xu and Sagaut [56] is adopted herein. With that,  $\sigma$  varies from 0 to a maximum value  $\sigma_0$  and its profile is formulated as

$$\sigma(\mathbf{x}) = \sigma_0 \frac{3125 (L_{\text{III}} - \mathbf{x} \cdot \mathbf{x}_0) (\mathbf{x} - \mathbf{x}_0)^4}{256 L_{\text{III}}^5}, \quad (36)$$

where  $L_{\text{III}}$  is the width of the non-reflective sound boundary layer and  $\mathbf{x}_0$  is the interface position. A higher  $\sigma_0$  is better at reducing reflection, but when it exceeds a critical value, notable reflection recurs [42]. In the present work,  $\sigma_0$  is set to  $10^{-3}$ .

#### 2.4. Numerical setup and algorithm pseudo-code

The body-fitted grid in the physical domain is depicted in Fig. 5. Setting the center of the circular cylinder as the origin, an O-type grid with  $N_\theta \times N_r$  is generated by stretching along the radial direction in polar coordinates. The radial distance  $r$  is from the origin and  $\theta$  is the counterclockwise angle from the positive direction of  $y = 0$ . The corresponding computational domain is a uniform Cartesian grid where  $N_\xi$  and  $N_\eta$  denote the number of lattice points in the  $\xi$  and  $\eta$  directions, respectively. The computation of the fluid dynamics and noise radiation have different grid resolution requirements. The flow structures convect downstream at a velocity close to the upstream velocity, but the sound waves propagate at the sound speed, so the acoustic wavelength is significantly larger than the characteristic length of flow dynamics. This difference in characteristic lengths leads to different grid resolutions. The grid adjacent to the cylinder should be fine enough to resolve the flow structures, while only coarse grids are needed to maintain low-dispersion and low-dissipation for sound waves in the far-field. Moreover, considering that the PML technique requires the non-reflective sound boundary layer to be larger than one acoustic wavelength, the computation domain is divided into three regions: (I) the flow region, (II) the sound region, and (III) the non-reflective buffer layer. In regions I and III, the grid node positions vary exponentially. In region II, the grid nodes are uniformly distributed along the radial direction. Taking  $N_\xi \times N_\eta = 512 \times 768$  as an example, the grid generation is shown in Fig. 6, and can be summarized as

$$\begin{aligned} \text{Region I (exponential)} : r_\eta &= 0.5 L_0 r_{v,\eta}, 0 < \eta < N_{\eta 1}, \\ \text{Region II (linear)} : r_\eta &= \Delta r_{v,N_{\eta 1}} (\eta - N_{\eta 1}) + 0.5 L_0 r_{v,N_{\eta 1}}, N_{\eta 1} < \eta < N_{\eta 2}, \\ \text{Region III (exponential)} : r_\eta &= 0.5 L_0 r_{v,\eta - N_{\eta 2} + N_{\eta 1}} + \Delta r_{v,N_{\eta 1}} (N_{\eta 2} - N_{\eta 1}), N_{\eta 2} < \eta < N_\eta, \end{aligned} \quad (37)$$



**Table 1**

Grid convergence study and validation for the scenario involving a fixed cylinder.  $p'_{rms}$  is measured at  $r = 80L_0$  and  $\theta = 90^\circ$ .

| Case                | No. of grid cells ( $N_\theta \times N_r$ ) | $C_D$ | $C_{L,rms}$ | $p'_{rms} (\times 10^{-5})$ | $St$   |
|---------------------|---------------------------------------------|-------|-------------|-----------------------------|--------|
| $120L_0 + 80L_0$    | $256 \times 384$                            | 1.301 | 0.355       | 7.650                       | 0.180  |
|                     | $512 \times 768$                            | 1.340 | 0.376       | 8.104                       | 0.184  |
|                     | $1024 \times 1536$                          | 1.349 | 0.380       | 8.165                       | 0.184  |
| $260L_0 + 130L_0$   | $512 \times 1280$                           | 1.339 | 0.375       | 8.138                       | 0.183  |
| Komatsu et al. [29] | –                                           | 1.332 | 0.3643      | 7.900                       | 0.1832 |
| Ma et al. [40]      | –                                           | 1.334 | 0.367       | –                           | 0.182  |

where  $r_{v,\eta} = e^{\eta \ln(1+2\pi/N_\xi)}$  describes the exponential distribution of the grid nodes. The characteristic length  $L_0$  is defined as the diameter of the cylinder. The corresponding borders of different regions in the computational domain,  $N_{\eta 1}$  and  $N_{\eta 2}$ , can be adjusted depending on the size of the physical domain. The radius of the physical domain is set to be about  $200L_0$ , and the thickness of the non-reflective sound boundary layer is about  $80L_0$ , which is in line with the requirements of the PML.

Grid resolution validation is carried out for three similar grids with different  $N_\xi \times N_\eta$ , which are specifically (a)  $256 \times 384$ , (b)  $512 \times 768$ , and (c)  $768 \times 11520$ . The CFL number is chosen to be a value close to 0.7 according to the grid. The Strouhal number ( $St$ ) is utilized to describe the period of vortex shedding and is defined as

$$St = f_v L_0 / U_0, \quad (38)$$

where  $f_v$  is the vortex shedding frequency and is counted for more than 30 periods using a fast Fourier transform (FFT). The fluid dynamic coefficient acting on the cylinder is normalized by

$$C_D = \frac{F_x}{\frac{1}{2} \rho_0 U_0^2 L_0}, \quad C_L = \frac{F_y}{\frac{1}{2} \rho_0 U_0^2 L_0}, \quad (39)$$

where  $C_D$  and  $C_L$  are the drag and lift coefficients, respectively.  $F_x$  and  $F_y$  are the force components, which are calculated from integration over the total stress tensor acting on the cylinder's surface in the computational domain and then transformed into the physical domain.  $Re$  is defined as  $Re = U_0 L_0 / \nu$  and set as  $Re = 150$ . By comparing the fluid dynamic coefficients in Table 1, it is found that there is little difference between the medium and finest grids, while the deviation in the results from the coarse grid is significant. Therefore, the grid with medium resolution is utilized in the present study, which is  $512 \times 768$ .

Furthermore, the effect of the thickness of the non-reflective sound boundary layer is also evaluated. By extending regions II and III in the medium grid, the physical domain becomes about  $390L_0$  with  $130L_0$  for the non-reflective buffer layer. The root-mean-square (RMS) of  $p'$  is denoted by  $p'_{rms}$  and calculated over 30 vortex shedding periods. The difference in  $p'_{rms}$  is less than 1%, which means  $80L_0$  for the non-reflective sound boundary layer is sufficient.

In terms of the number of grid nodes, the GICLBM can significantly reduce memory consumption compared to the LBM on a uniform grid [31], as evidenced in Table 1. Moreover, the extra interpolation operations introduced in the streaming procedure indeed worsen the computational efficiency, if compared with the conventional LBM. However, thanks to the grid used here being much more efficient than the LBM with a uniform grid, the overall computational time can be remarkably reduced.

The pseudo-code for the GILBM is shown in Algorithm 1. For post-processing the results in the absolute frame of reference, the transformation from the moving reference of frame is essential. The information in the non-reflective sound boundary layer is usually not of concern, and the interpolation is carried out in regions I and II. It should be noted that the coordinates of stationary observer points can be quickly converted from the physical domain to the computational domain due to the grid being algebraically generated. Using the bilinear interpolation, all data in the moving frame of reference are transformed into the absolute frame of reference.

### 3. Results and discussions

#### 3.1. Scenario I: a fixed circular cylinder

The GICLBM is first applied to a classical two-dimensional flow past a fixed circular cylinder. As noise generation is closely related to the flow field, results for the fluid dynamics obtained from the GICLBM are first verified. Periodic vortex shedding occurs when  $Re$  is larger than a critical value of about 47, and the fluid dynamics of the cylinder surface also vary periodically. Hence,  $Re = 150$  is selected for validation. Under this condition, the flow field is still in the laminar regime and the frequencies of fluid dynamic and the far-field fluctuating pressure are related to the vortex shedding frequency. The instantaneous vorticity contour is depicted in Fig. 7(a), and the positive and negative vortices are alternately shed from the cylinder. The vorticity is defined as  $\omega = 0.5 (\partial_x u_y - \partial_y u_x)$  and is transformed from the computational domain into the physical domain as

$$\begin{bmatrix} \partial_x u_x & \partial_y u_x \\ \partial_x u_y & \partial_y u_y \end{bmatrix} = \begin{bmatrix} \partial_\xi u_\xi & \partial_\eta u_\xi \\ \partial_\xi u_\eta & \partial_\eta u_\eta \end{bmatrix} \begin{bmatrix} \partial_\eta y & -\partial_\eta x \\ -\partial_\xi y & \partial_\xi x \end{bmatrix} J^{-1}. \quad (40)$$

Similarly, the vortex can be also identified by  $\lambda_{ci}$  criterion [59]. The time-averaged pressure coefficient is defined as

**Algorithm 1** Pseudo-code of the GICLBM for fluid-structure-sound interaction.**Initialization**

```

1: Allocate memory for all variables,  $f, f^*, u, p, x, \bar{e}, \Delta\xi, J, Q$ 
2: for each grid cell do
3:   Grid partition so as to determine  $x$  and  $J$ 
4:   Calculate the contravariant velocities  $\bar{e}$  in Eq. (17)
5:   Determine global time step size  $\Delta t$  with the CFL constraint in Eq. (24)
6:   Calculate  $\Delta\xi[k]$  using the two-step Runge-Kutta scheme, i.e., Eq. (19)
7:   Initialize variables  $f, f^*, u, p, x, \bar{e}, \Delta\xi, J, Q$ 
8: end for

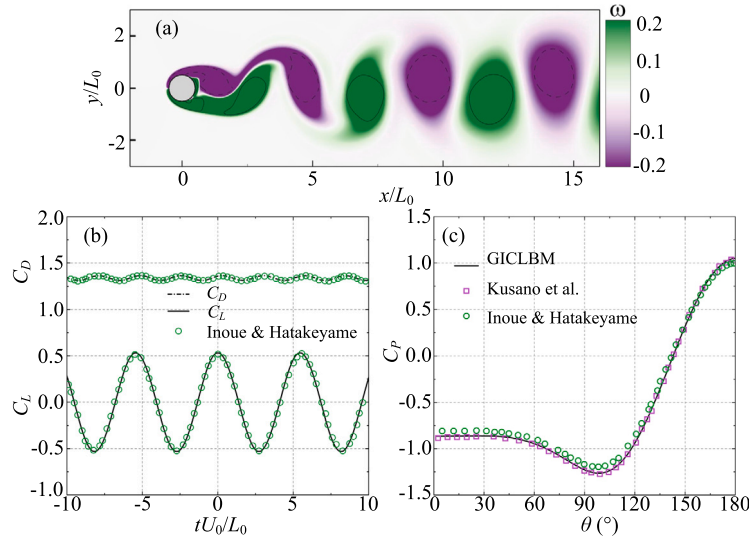
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**Time marching**

```

1: while Timestep < MaxTimestep do
2:   for each grid cell do
3:     Calculate  $\rho, u$  using Eq. (4)
4:     Update  $Q$  and calculate  $\Omega^{PML}$  using Eq. (34)
5:     Calculate the body force terms  $\bar{F}$  by Eq. (30)
6:     Execute the collision step locally by Eq. (16)
7:     Implement the PML in collision step by adding  $\Omega^{PML}$  in Eq. (33)
8:   end for
9:   for each boundary grid cell do
10:    Determine the dynamic boundary condition using Eq. (32)
11:    Enforce boundary conditions via the non-equilibrium extrapolation scheme
12:   end for
13:   for each grid cell do
14:    Perform streaming step using interpolation  $f^*(\xi - \Delta\xi, t)$  from  $f(\xi, t)$ 
15:    Update  $f^*$  with  $f$ 
16:   end for
17:   Solve the kinematic dynamics using Eq. (31)
18:   Output flow field quantities or monitored variables if needed
19:   Timestep++
20: end while

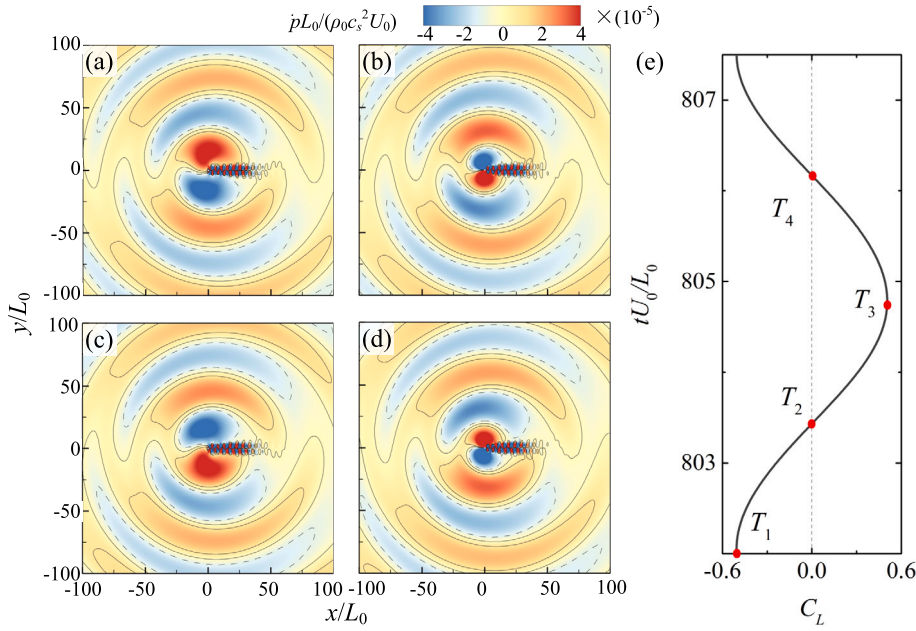
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**Fig. 7.** With  $Re = 150$  and  $Ma = 0.2$ , (a) instantaneous vorticity contours, (b) the lift coefficient  $C_L$  and the drag coefficient  $C_D$ , and (c) the time-averaged pressure coefficient  $C_P$  in comparison with previous references [25,30], shown as green circles and purple squares, respectively. The positive and negative vorticities are enclosed by  $\lambda_{ei} = \pm 0.2$  which are denoted by solid and dashed isolines, respectively. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

$$C_P = \frac{\bar{p}}{\frac{1}{2}\rho_0 U_0^2}. \quad (41)$$

As is shown in Fig. 7(b), the instantaneous  $C_D$  and  $C_L$  calculated from the GICLBM are both sinusoidal curves and in good agreement with the results from Inoue and Hatakeyama [25], which were obtained using finite difference methods. The drag is composed of the pressure difference caused by the flow separation and the friction force, and thus is subject to the flow direction. The symmetric vortex shedding gives an average  $C_L$  of 0. Meanwhile, the frequency of  $C_D$  is double that of  $C_L$ . The time-averaged pressure coefficient  $C_P$  in Fig. 7(c) shows a similar trend to the results from Inoue and Hatakeyama [25], but slightly lower in the leeward direction. Another result calculated from the LBM [30] coincides with that from the GICLBM.



**Fig. 8.** Instantaneous dilatation contours at four typical times: (a)  $T_1 = 802.05$ , (b)  $T_2 = 803.43$ , (c)  $T_3 = 804.79$ , and (d)  $T_4 = 806.17$ , which are shown in (e), with  $Re = 150$  and  $Ma = 0.1$ . The sound waves are denoted by solid and dashed isolines with values of  $\pm 1 \times 10^{-5}$ .

Inoue and Hatakeyama [25] demonstrated that periodic vortex shedding leads to sound generation from a circular cylinder. In general, noise is denoted by a fluctuating pressure  $p' = p - \bar{p}$ , where  $\bar{p}$  is the mean pressure. Similarly, the dilatation  $\dot{p} = \partial p / \partial t$  can be used to evaluate the noise without time averaging and can be normalized using  $(\rho_0 c_s^2 U_0) / L_0$ . The instantaneous dilatation contours at four typical times are depicted in Fig. 8. As shown by Inoue and Hatakeyama [25], the positive and negative sound waves are alternately generated from the cylinder and the lift dipole is dominant. The relationship between the lift and sound waves is shown in Fig. 8, where  $T_1$ ,  $T_2$ ,  $T_3$ , and  $T_4$  are the peak, trough, and equilibrium positions of the lift curve, respectively. From  $T_1$  to  $T_3$ , as the lift decreases, a negative sound wave is generated from the upper side and a positive sound wave radiates from the lower. On the contrary, with the increase in the lift force after  $T_3$ , a positive sound wave appears on the upper side and a negative sound wave appears on the lower. The pressure fluctuation is induced from the surface of the cylinder due to vortex shedding and radiates to the far-field as a sound wave. This mechanism was summarized by Inoue and Hatakeyama [25] as: when a vortex is shed from one side of the cylinder, a negative pressure pulse is induced on the same side and a positive pressure pulse on the opposite side. These are then radiated to the far-field.

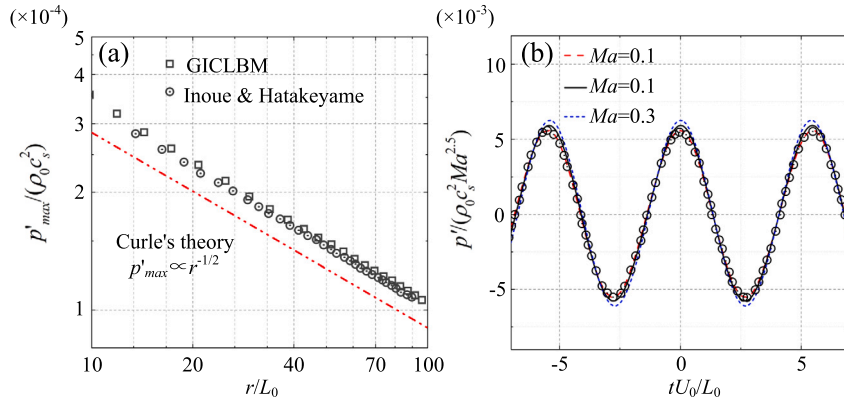
Inoue and Hatakeyama [25] provided several benchmarks for acoustic computations of a flow past a fixed circular cylinder and derived the two-dimensional Curle's solution to describe the propagation of noise. By substituting the two-dimensional Green's function, Curle's theory is supplemented with two-dimensional noise radiated from the solid boundary. In particular, assuming the solid structure is compact and ignoring the effect of a quadrupole due to a relatively low  $Ma$ , Curle's solution can be expressed as

$$p'(\mathbf{x}, t) = \frac{\vec{r}}{|\mathbf{r}| \pi \sqrt{8} |\mathbf{r}| c_s - \infty} \int_0^t \frac{\partial \mathbf{F}(\mathbf{x}, \tau + \tau')}{\partial \tau} \frac{d\tau'}{\sqrt{-\tau'}}, \quad (42)$$

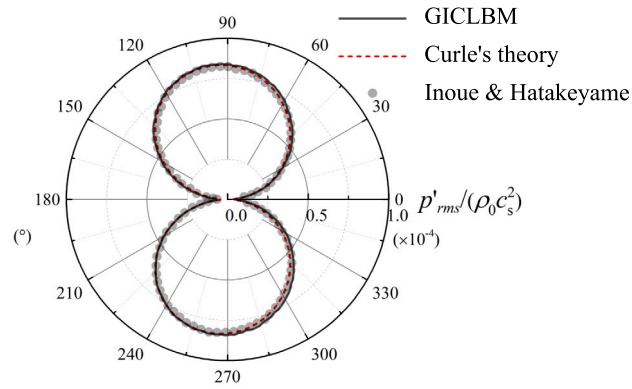
where  $\tau = t - |\mathbf{r}|/c_s$  is the retarded time. Due to the cylinder being considered compact and therefore equivalent to a sound source located at the origin of the polar coordinates, the distance  $\mathbf{r}$  is equivalent to the vector from the sound source position to the observer and  $\tau' = \tau' + \tau$  represents the effect of sound waves from previous points in time. The letter  $\mathbf{F}$  is the force acting on the cylinder. The sound pressure at the observer can be obtained by spatial and temporal integration.

$Ma$  is another similarity criterion in the CAA, and is defined as  $Ma = U_0/c_s$ . For  $Ma = 0.2$ , the observer points are arranged along  $\theta = 90^\circ$ . In Fig. 9(a), the maximum sound pressure  $p'_{max}$  theoretically declines with distance and satisfies  $1/\sqrt{r}$ , which is consistent with the findings of Inoue and Hatakeyama [25]. Moreover, as is shown in Fig. 9(b), the instantaneous  $p'$  measured at  $(0, 75L_0)$  is also in agreement. The instantaneous sound pressure curve is a single sinusoidal curve with a frequency consistent with the lift curve.

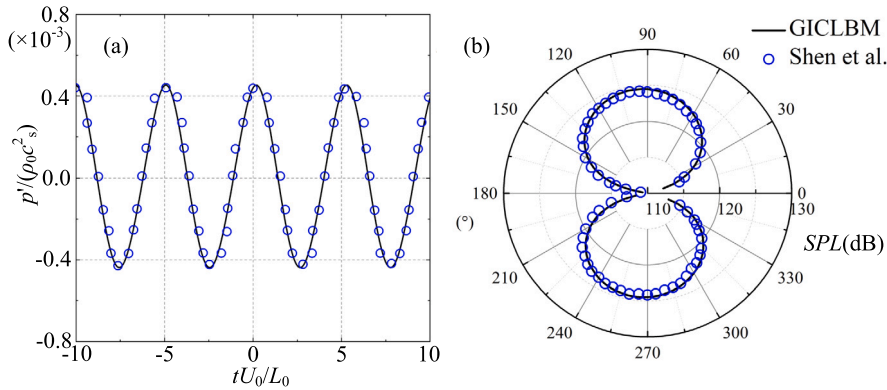
Finally, the acoustic radiation directivity is depicted in polar coordinates. Fig. 10 shows the  $p'_{rms}$  measured at  $r = 75L_0$  with equal angle intervals. It should be noted that the results are obtained by bilinear interpolation although the cylinder is stationary. As expected, the present results are slightly larger than the results obtained by Inoue and Hatakeyama [25]. The reason for this is that  $C_L$  in the present work is larger, which leads to a more intense lift dipole radiating noise. Furthermore, the theoretical values calculated from the two-dimensional Curle's solution [25] are in good agreement with the GICLBM.



**Fig. 9.** Validation of the acoustic results at  $Re = 150$ . (a) The decay of the maximum fluctuating pressure along  $\theta = 90^\circ$ , where squares represent the present results and circles are the results from Inoue and Hatakeyama [25] at  $Ma = 0.2$ . The red dash-dot-dot line shows the theoretical decay trend, which is proportional to  $1/\sqrt{r}$ . (b) The instantaneous fluctuating pressure measured at  $(0, 75L_0)$  for  $Ma = 0.1, 0.2$ , and  $0.3$ , is denoted by red dashed, solid, and short dashed lines, respectively.



**Fig. 10.** Validation of the acoustic radiation directivity measured at  $r = 75L_0$  in  $Re = 150$  and  $Ma = 0.2$ . The present direct simulation result is denoted by the black solid line. The solution calculated from two-dimensional Curle's theory based on the present fluid dynamic data is denoted by the red short-dashed line. The direct simulation results from Inoue and Hatakeyama [25] are denoted by the solid gray circle.



**Fig. 11.** A comparison of the instantaneous fluctuating pressure at  $(0, 75L_0)$  and the acoustic radiation directivity measured at  $r = 75L_0$  for the present results (black solid line) and the previous reference [48] (blue circles).

The sound generation for a two-dimensional circular cylinder at  $Re = 200$  and  $Ma = 0.2$  is also compared. The sound pressure level (SPL) is defined as  $SPL = 20 \log_{10} \left( (p' / \rho_0 c_s^2) / p_{ref}^* \right)$ , where  $p_{ref}^*$  is set as  $2 \times 10^{-5}$  Pa and  $p_{ref}^*$  is normalized by a realistic air density and sound speed. Fig. 11 shows the instantaneous fluctuating pressure and acoustic radiation directivity are in good agreement with the results of Shen et al. [48], where the finite volume method was utilized. The above validation shows that the GICLBM can accurately simulate the sound generation and propagation from a fixed cylinder. It should be noted that the observer points downstream of the cylinder are not displayed because their noise may be polluted by vorticity.

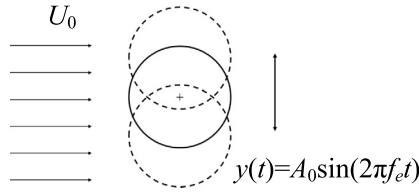


Fig. 12. A schematic diagram of a forced oscillating circular cylinder.

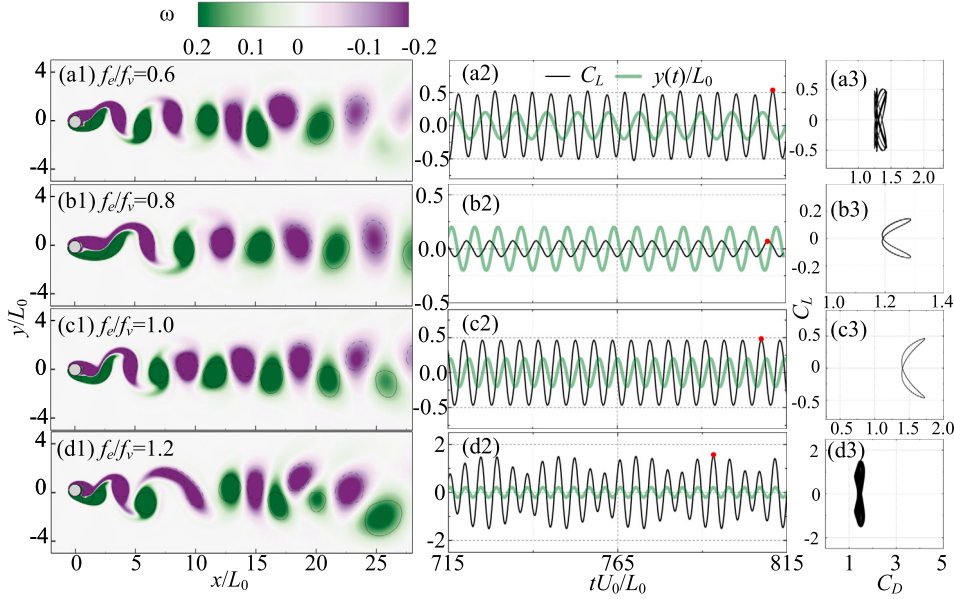


Fig. 13. Instantaneous vorticity contours, the lift coefficient  $C_L$ , the cylinder's displacement  $y(t)$  curves, and the  $C_L - C_D$  phase diagrams for different oscillating frequencies: (a)  $f_e/f_v = 0.6$ , (b)  $f_e/f_v = 0.8$ , (c)  $f_e/f_v = 1.0$ , and (d)  $f_e/f_v = 1.2$  with  $Re = 150$  and  $Ma = 0.2$ . The positive and negative vorticities are enclosed by  $\lambda_{ci} = 0.1$ , and the solid and dashed lines represent the regions with positive vorticity and negative vorticity, respectively. The bold solid green line is  $y(t)$ , the black line is  $C_L$ , and the red circle is the moment of the vorticity contour, which is the maximum  $C_L$ .

### 3.2. Scenario II: a forced oscillating circular cylinder

In this section, the case in which a circular cylinder is forced to oscillate is considered to verify the accuracy of the present algorithm for a moving rigid body. The previous grid generated for a fixed circular cylinder will continue to be utilized.  $Re$  is set to 150 and  $Ma$  is set to 0.2.

The circular cylinder is forced to transversely oscillate periodically, as is shown in Fig. 12, and the displacement  $y(t)$  can be described as

$$y(t) = A_0 \sin(2\pi f_e t), \quad (43)$$

where  $A_0$  and  $f_e$  are the oscillating amplitude and frequency, respectively. In the present work, the amplitude  $A_0$  is set as  $A_0 = 0.2L_0$  and  $f_e$  varies from  $0.2 f_v$  to  $1.4 f_v$ .

As indicated by previous works [40,55], different vortex structures appear in this case, and both the fluctuating pressure and acoustic radiation directivity in the far-field are significantly altered when the cylinder is undergoing forced oscillation. Fig. 13 shows the instantaneous vorticity contours under different  $f_e$ . The positive and negative vortices alternately shed from the cylinder and migrate downstream. For  $f_e/f_v = 0.6$ , the wake pattern is similar to that of the fixed cylinder in Fig. 7. At  $f_e/f_v = 0.8$ , the shear layer is elongated and then decreases as the oscillating frequency increases. Moreover, at  $f_e/f_v = 1.2$ , a switch in the vortex is found in the adjacent region relative to the displacement of the cylinder, which is also consistent with the findings of Ma et al. [40]. In the downstream location away from the cylinder, the vortex is in a chaotic state when  $f_e/f_v = 0.6$  and  $1.2$ . The instantaneous  $C_L$  and  $y(t)$  curves are also depicted in Fig. 13. It can be found that the lift force changes periodically with the displacement of the cylinder at  $f_e/f_v = 0.8$  and  $f_e/f_v = 1.0$ . This phenomenon is also called the lock-on regime. However, at  $f_e/f_v = 0.6$ , the period of  $C_L$  is similar to that of the fixed cylinder without control but smaller than that of the cylinder displacement. At  $f_e/f_v = 1.2$ , the lift curve is not a single sinusoidal curve, indicating the existence of harmonics caused by nonlinear effects from FIV. In the lock-on regime, the vortex shedding frequency is consistent with  $C_L$ , deviating from that of a fixed cylinder and turning into an oscillating frequency. The length of the shear layer also changes with the vortex shedding period. However, when the lock-on regime does not occur, the



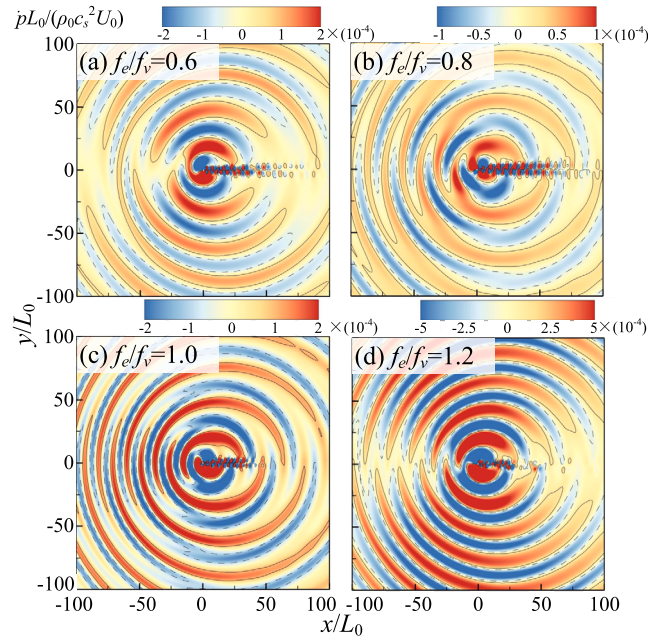


Fig. 14. Instantaneous dilatation contours for different oscillating frequencies: (a)  $f_e/f_v = 0.6$ , (b)  $f_e/f_v = 0.8$ , (c)  $f_e/f_v = 1.0$ , and (d)  $f_e/f_v = 1.2$  for  $Re = 150$  and  $Ma = 0.2$ . The sound waves are denoted by solid and dashed isolines with different values: (a)  $\pm 5 \times 10^{-5}$ , (b)  $\pm 2 \times 10^{-5}$ , (c)  $\pm 1 \times 10^{-4}$ , and (d)  $\pm 1 \times 10^{-4}$ .

vortex shedding is always modulated by cylinder vibration and does not converge to a stable state, which eventually leads to chaotic wakes and disorganized  $C_L - C_D$  diagram.

The instantaneous noise radiated from the forced oscillating circular cylinder is depicted in terms of the dilatation in Fig. 14. It can be seen that the overall noise radiation is augmented with increasing oscillating frequency. The sound waves are denoted by the solid and dotted lines. When the oscillating frequency is lower, as is shown in Fig. 14. (a) when  $f_e/f_v = 0.6$ , the noise radiated from the forced oscillating cylinder is similar to that from the fixed cylinder, and dominated by lift dipoles in the far-field. In the lock-on regime shown in Fig. 14 (b)  $f_e/f_v = 0.8$  and (c)  $f_e/f_v = 1.0$ , the wavelength of the sound wave decreases with the increase in the oscillating frequency, and the drag dipole is enhanced. The acoustic radiation directivity is affected by both the lift and drag dipole, which leads to a difference from the fixed cylinder. Meanwhile, at  $f_e/f_v = 1.2$ , which is outside of the lock-on regime, the acoustic radiation directivity is similar to that of the fixed cylinder and dominated by the lift dipole. In addition, the wavelength of its sound waves gets shorter. The differences in the fluid dynamics and noise radiation both indicate that the forced oscillating circular cylinder is more complicated and a further severe challenge for the present algorithm.

The fluid dynamic coefficient is first verified in Fig. 15. The trends of the RMS values of the lift coefficient  $C_{L,rms}$  and the average drag coefficient  $\bar{C}_D$  calculated from the GICLBM are consistent with the results from Ma et al. [40]. In Fig. 15, the lift coefficient decreases as  $f_e$  increases before  $f_e/f_v = 1.0$ , and then increases when  $f_e/f_v > 1.0$  in comparison with the fixed cylinder. Under the oscillating regime, the drag coefficients almost entirely increase. Moreover, the variation in  $C_L$  corresponds to the sound pressure in the lift direction, which indicates that forced oscillation is an effective way to reduce the radiation noise in the lift direction by suppressing the lift fluctuation.

The acoustic radiation directivity is depicted in polar coordinates and compared with the results of Ma et al. [40], which were calculated using the finite volume method. Considering the Doppler effect on sound propagation,  $p'_{rms}$  is measured at  $r = r'(1 + Ma \cos \theta)$  with  $r' = 75L_0$ . As is shown in Fig. 16, the present results are in agreement with those of Ma et al. [40]. In particular, the quadrupole characteristic in Fig. 16(b) is captured. The MRT-LBM is also evaluated and compared. Due to the bulk viscosity in the MRT-LBM approaching that of the GICLBM, the results for the far-field noise are similar.

In addition, the instantaneous curve of  $p'$  measured at  $(0, 80L_0)$  is also consistent with the results of Komatsu et al. [29] at  $Re = 150$  and  $Ma = 0.2$ , which were calculated using the finite difference method. The acoustic radiation directivity is shown in terms of  $p'_{rms}$  at  $r = 80L_0$ . The noise radiation shows a lift dipole characteristic and is also in agreement with the results of Komatsu et al. [29]. In Fig. 17, the  $p'_{rms}$  near  $\theta = 60^\circ$  is slightly larger than the results of Komatsu et al. [29] but close to Zhao et al. [58] which was calculated by hybrid method.

The above numerical cases and results validate that the GICLBM coupled with the moving frame of reference can successfully resolve the flow dynamics of the FIV situation and the associated noise radiation.

### 3.3. Scenario III: a circular cylinder undergoing vortex-induced vibration

In this section, the GICLBM is utilized to investigate VIV. The circular cylinder is induced to vibrate by an alternately shedding vortex when  $Re$  is larger than a sub-critical value ( $Re \approx 20$  [8]). As is shown in Fig. 18, the initial position of the cylinder's center

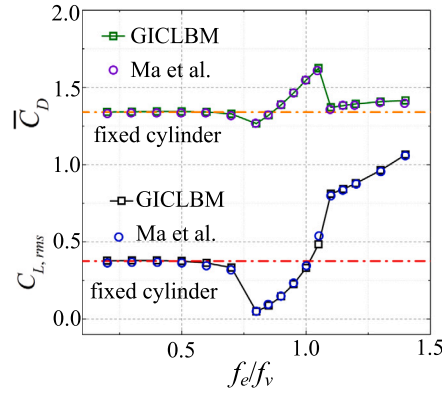


Fig. 15. A comparison of the present results for the fluid dynamic coefficient vs. the oscillating frequency  $f_e$  (squares) with the results for a fixed cylinder (dash-dotted lines) and from Ma et al. [40] (blue circles) with  $Re = 150$  and  $Ma = 0.2$ .

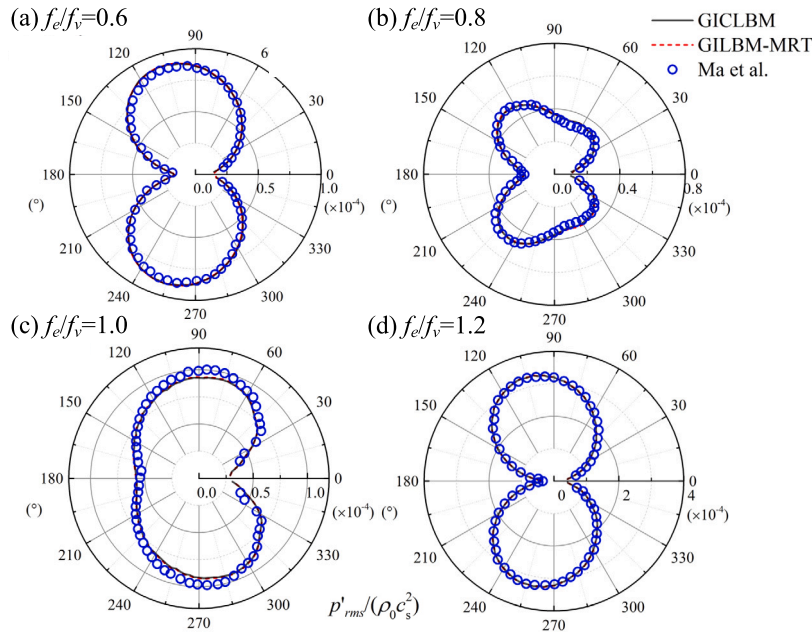


Fig. 16. A comparison of the acoustic radiation directivity depicted for  $p'_{rms}$  measured at  $r' = 75L_0$  at oscillating frequencies (a)  $f_e/f_v = 0.6$ , (b)  $f_e/f_v = 0.8$ , (c)  $f_e/f_v = 1.0$ , and (d)  $f_e/f_v = 1.2$  (solid black lines), with the results from Ma et al. [40] shown as blue circles for  $Re = 150$  and  $Ma = 0.2$ .

is set as the origin, and the displacement of the cylinder  $y(t)$  in the cross-flow direction is described by a mass-spring system with a single degree of freedom,

$$m\ddot{y} + Ky = F_L, \quad (44)$$

where  $m$  is the mass of the cylinder,  $K$  is the spring stiffness, and  $F_L$  is the lift force from the external fluid. The non-dimensional reduced velocity  $U_r$  is defined as

$$U_r = \frac{U_0}{f_n L_0}, \quad (45)$$

where  $f_n$  is the natural frequency of the structure and is calculated as

$$f_n = \frac{1}{2\pi} \sqrt{\frac{K}{m}}. \quad (46)$$

The non-dimensional mass ratio is defined as

$$m^* = \frac{m}{\rho_0 L_0^2}. \quad (47)$$



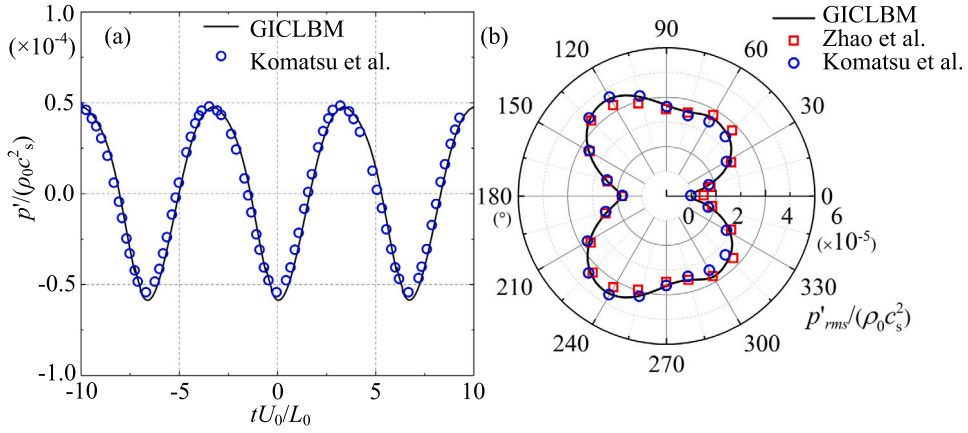


Fig. 17. A comparison of (a) the instantaneous fluctuating pressure measured at  $(0, 80L_0)$  and (b) the acoustic radiation directivity measured at  $r = 80L_0$  with the results of Komatsu et al. [29] and Zhao et al. [58] (black solid line, blue circles and red squares, respectively). The cylinder is forced oscillating with  $f_e = 0.15$  at  $Re = 150$  and  $Ma = 0.2$ .

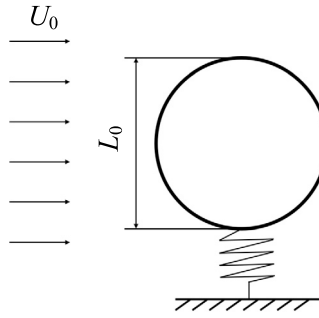


Fig. 18. A schematic diagram of a circular cylinder undergoing VIV.

In the present work,  $m^*$  is fixed to 2. The validation of the GICLBM on VIV is carried out by comparing the vibrating amplitude  $A^*$ , which is normalized by  $L_0$ , with other references at several  $U_r$ .

The similarity of  $Ma$  plays a key role in the CAA, and the VIV is first investigated for different  $Ma$  considering the effect of compressibility. As is depicted in Fig. 19, the trend of the maximum amplitude  $A^*$  with  $U_r$  is basically consistent. A significant jump in the vibrating amplitude occurs at  $U_r = 4$ . When the frequency of vortex shedding approaches  $f_n$ , the lock-in regime occurs. Due to being associated with a larger vibrating amplitude, the lock-in regime is usually of more concern. The present results are close to those of Jiang et al. [27] and Wang et al. [53]. At  $Ma = 0.2$ , the maximum differences in the lock-in regime are 5.5% and 4.7% when compared with the results of Jiang et al. [27] and Wang et al. [53], respectively. However, the differences decrease with  $Ma = 0.1$  to 2.1% and 1.2%, respectively. Although all of the above results are calculated based on the LBM, different treatments on the curved solid boundary may lead to acceptable deviations. Moreover, the results from the LBM are relatively far from those of Borazjani et al. [7], which were obtained by solving the unsteady incompressible NS equations.

The maximum difference in  $A^*$  between  $Ma = 0.1$  and  $Ma = 0.2$  is 4.8%, which indicates that  $Ma$  may influence VIV. Furthermore,  $A^*$ , the maximum overall force in the  $y$  direction,  $C_{y,max}$ , and  $St$  are calculated for various  $Ma = 0.01, 0.03, 0.05, 0.1, 0.2$  and  $0.3$ . The results are depicted in Fig. 20 at  $m^* = 2$  and  $U_r = 5$  when the cylinder is in the lock-in regime. It should be noted that the overall force in the  $y$  direction  $C_{y,max}$  is calculated as

$$C_y = \frac{(F_y - Ky)}{\frac{1}{2}\rho_0 U_0^2 L_0}. \quad (48)$$

As is shown in Fig. 20 (c), Mach number has little effect on the frequency of vortex shedding in the lock-in regime. However, the vibrating amplitude and fluid dynamics vary more and more significantly with increasing  $Ma$ . In terms of the vibrating amplitude, the difference in the results from the GICLBM with  $Ma = 0.01$  compared with those of Wang et al. [53] and Ren et al. [46] are 1.8% and 5.0%, respectively, but 11.5% away from  $Ma = 0.3$ . Similarly, the difference in the fluid dynamic coefficient is 13.9% between  $Ma = 0.01$  and  $Ma = 0.3$ . Hence, there may be a critical value of  $Ma$  for VIV problems. When exceeding  $Ma = 0.1$  in the present work, the vibration and fluid dynamics differ significantly, which may lead to a significant disparity in the far-field noise. Due to the previous cases of fixed and forced oscillating cylinders being verified for  $Ma = 0.2$  and  $0.3$ , respectively, it is reasonable to confirm that the present results are reliable.

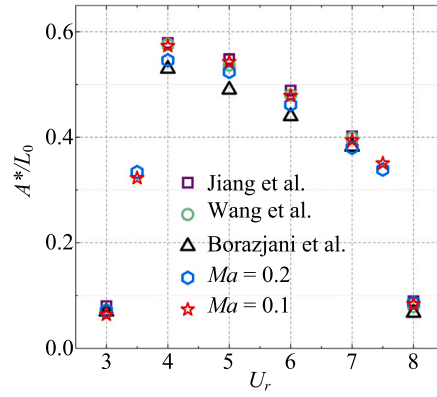


Fig. 19. The trend of the maximum amplitude  $A^*$  vs  $U_r = 3, 4, 5, 6, 7$ , and  $8$  at  $Re = 150$ . The purple squares, green circles, and black triangles denote the results from Jiang et al. [27], Wang et al. [53], and Borazjani et al. [7], respectively. The red stars and blue hexagons are calculated for  $Ma = 0.1$  and  $Ma = 0.2$ , respectively.

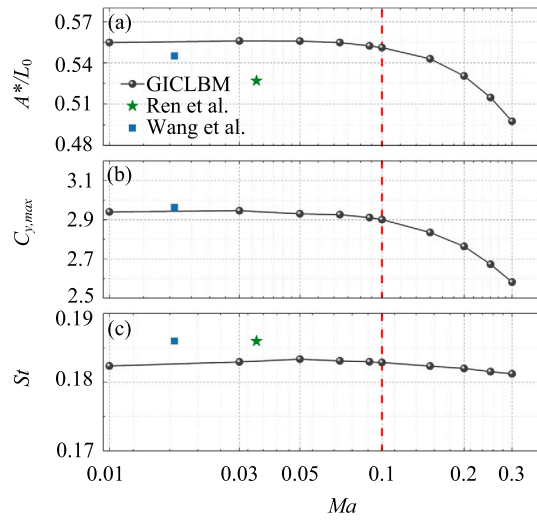
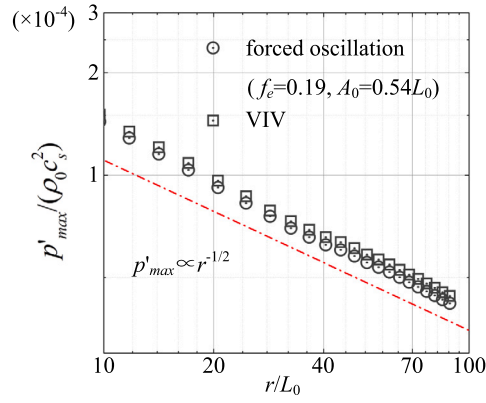


Fig. 20. For different Mach numbers,  $Ma = 0.01, 0.03, 0.05, 0.1$  and  $0.2$ , the trend of (a) the maximum vibrating amplitude,  $A^*$ , (b) the maximum overall force in the  $y$  direction,  $C_{y,max}$ , and (c) the Strouhal number,  $St$ , at  $Re = 100$ . The black solid circles represent the present results and the red dashed lines represent  $Ma = 0.1$ . The green stars and blue squares are from the results of Wang et al. [53] and Ren et al. [46], respectively.

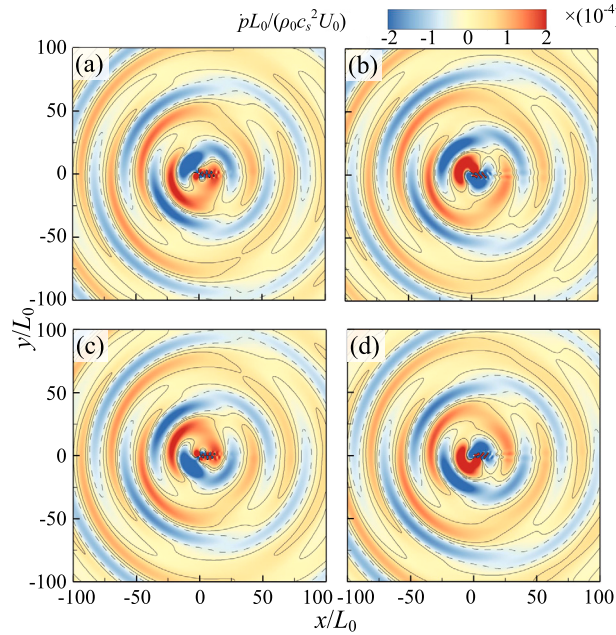
Based on the above analysis, the far-field noise is calculated for  $Ma = 0.1$ . Firstly, the acoustic results from GICLBM are verified using Curle's theory. To avoid the effect of the varying distance between the observer points and the sound source due to the transverse vibration, the observer points are arranged along  $\theta = 180^\circ$ . Moreover, based on the similarities between forced oscillation and VIV, an oscillating cylinder with  $A_0 = 0.54L_0$  and  $f_e = 0.19$  (approaching the vibrating parameters of VIV at  $m^* = 2$ ,  $U_r = 5$ ) is also considered. As is shown in Fig. 21, the maximum sound pressures  $p'_{max}$  are close and theoretically decline with distance as  $1/\sqrt{r}$ .

The instantaneous sound radiated from a vortex-induced vibrating cylinder is represented by dilatation and depicted in Fig. 22. Four typical moments are selected, which correspond to the peak, trough, and equilibrium positions ( $y = 0$ ) in the displacement curve, as is shown in Fig. 23(d). Besides the direction of the lift, sound waves also generate and propagate along the direction of the drag, which is similar to Fig. 14 (b) and (c).

In Fig. 22, as the cylinder moves from the (a) peak to (b) equilibrium position, a positive sound wave radiates from the cylinder along approximately  $\theta = 135^\circ$ . A negative sound wave then radiates along about  $\theta = 225^\circ$  when the cylinder moves from (b) the equilibrium position to (c) the trough. In contrast, when the cylinder moves along the positive direction ( $y > 0$ ), the sound wave radiates in the opposite state. When the cylinder moves from (c) the trough to (d) the equilibrium position, a positive sound wave is generated on the lower side of the cylinder. As the cylinder moves from (d) the equilibrium position to (a) the peak, a negative sound wave appears on the upper side. The instantaneous vorticity contours at the moment of maximum lift coefficient are depicted in Fig. 23(a). The wake patterns of the fixed and forced oscillating cylinders are similar. Positive and negative vortices alternately shed in what is called a "2S" pattern. But this differs for VIV, where the vortices separate from the centerline ( $y = 0$ ) and migrate downstream in what is called a "C2S" pattern. The instantaneous fluid dynamic curves are shown in Fig. 23 (c). The frequency of the drag coefficient  $C_D$  is twice that of the lift force, and significantly increases when the cylinder is in movement. This result explains the enhancement of the sound waves in the drag direction in Fig. 22. The lift curve of a fixed cylinder is a single sine wave, but becomes



**Fig. 21.** The decay of the maximum fluctuating pressure along  $\theta = 180^\circ$  at  $Re = 150$  and  $Ma = 0.1$ . The squares represent the cylinder in the VIV regime at  $m^* = 2$ ,  $U_r = 5$ , and the circles represent forced oscillation with  $A_0 = 0.54L_0$  and  $f_e = 0.19$ . The red dash-dot-dot line is the theoretical decay trend, which is proportional to  $1/\sqrt{r}$ .

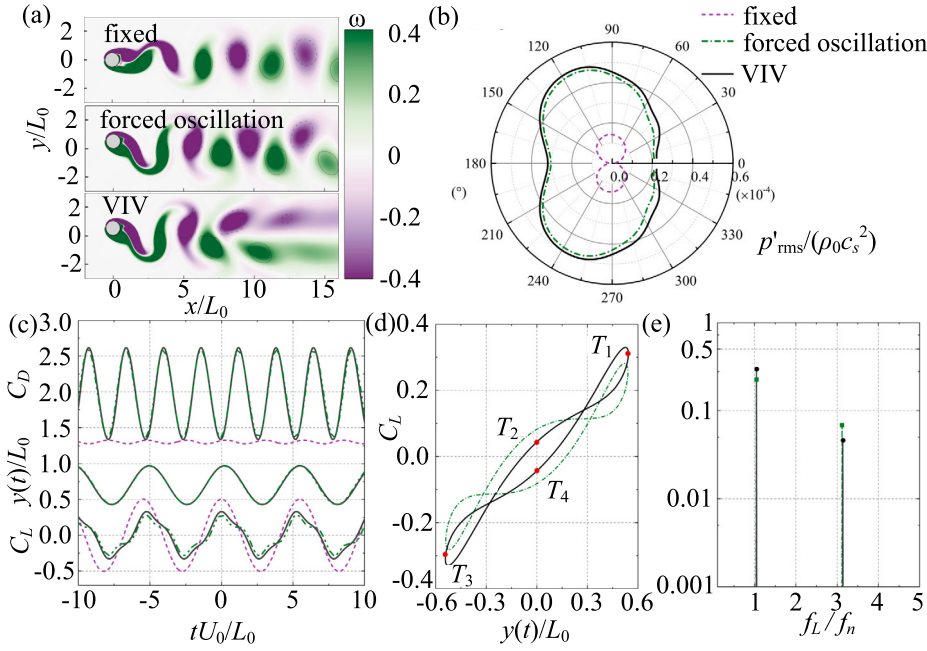


**Fig. 22.** Instantaneous dilatation contours at four typical times: (a)  $T_1 = 800.87$ , (b)  $T_2 = 802.17$ , (c)  $T_3 = 803.49$ , and (d)  $T_4 = 804.78$  when the cylinder is undergoing VIV. The sound waves are denoted by solid and dashed isolines,  $\pm 5 \times 10^{-5}$ .

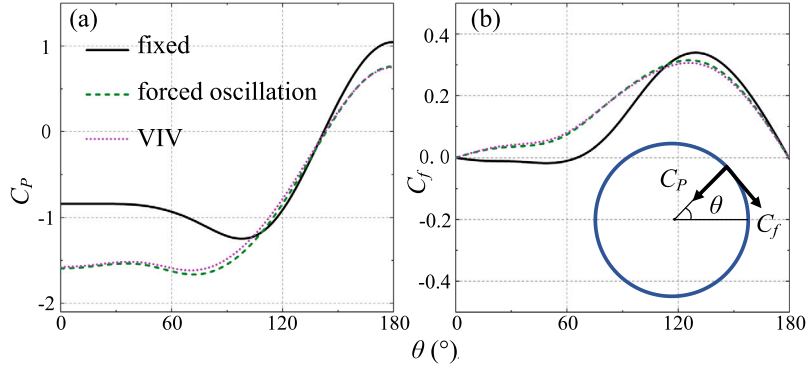
complicated when the cylinder moves. FFT is also implemented for more than 30 vortex-shedding periods to analyze the frequency spectrum. In Fig. 23(e), results show that the harmonics are enhanced due to the movement of the cylinder, and the harmonics of the forced oscillation are stronger than those of the VIV. Fig. 23(d) depicts the  $C_L - A^*$  phase diagram, showing both clear and closed curves. It can be found that the positions of the lift extreme values do not coincide with that of the displacement for both forced oscillation and VIV. Similar stress distributions can be observed on the surface of moving cylinders, but these distributions change when the cylinder is fixed. The pressure coefficient on the leeward side is still lower than the windward, and is lower than 1 at stagnation point. The separation point of the fixed cylinder is located at about  $\theta = 65^\circ$  where the time-averaged skin-friction coefficient  $C_f$  is 0. When the cylinder is moving, the skin-friction coefficient is greater than 0, indicating that the separation point's location cannot be determined from the time-averaged wall stress distribution.

The acoustic radiation directivity of the cylinder is represented by  $p'_{rms}$  measured at  $r' = 75L_0$  and depicted in Fig. 23(b). It can be found that although the maximum lift of the fixed cylinder is larger than that of the other two regimes, the noise in the lift direction is significantly smaller in contrast. This indicates that the periodic movement of the cylinder generates some other sound sources. Moreover, the acoustic radiative directivity of the VIV in Fig. 24 is not exactly the same as that of the forced oscillating cylinder, especially at positions near  $\theta = 120^\circ$  and  $\theta = 60^\circ$ .

In order to investigate the sound sources from VIV, the fluctuating pressure can be decomposed into monopole, dipole, quadrupole, and other sound sources. This decomposition was demonstrated by Inoue and Hatakeyama [25], and has been utilized for acoustic



**Fig. 23.** A comparison of the fixed, forced oscillating, and vortex-induced vibrating cylinders in terms of the fluid dynamic and noise radiation at  $Re = 150$  and  $Ma = 0.1$ . (a) Instantaneous vorticity contours, (b) the acoustic radiation directivity measured at  $r' = 75L_0$ , (c) the instantaneous lift coefficient  $C_L$ , the drag coefficient  $C_D$  and  $(0.5A^* + 0.7)$ , (d) the  $C_L - A^*$  phase diagram and (e) the FFT results for  $C_L$ . The positive and negative vorticities are enclosed by  $\lambda_{ci} = 0.2$  isolines, and the solid and dashed isolines represent zones with positive and negative vorticity, respectively. The time  $T_1$ ,  $T_2$ ,  $T_3$ , and  $T_4$  correspond to the peak, trough, and equilibrium positions ( $y = 0$ ) of the displacement curve, respectively, which are denoted by red solid circles in (c).



**Fig. 24.** A comparison of the fixed, forced oscillating, and vortex-induced vibrating cylinders in terms of the time-averaged pressure coefficient  $C_p$  and skin-friction coefficient  $C_f$ .

research [40]. The details of the composition are provided in the appendix. Since the decomposition results can be polluted by the vortex, the observer points should be far from the wake and are located at  $r' = 75L_0$  in the present work. In Fig. 25, three positions at  $\theta = 45^\circ, 90^\circ$ , and  $180^\circ$  are exhibited, corresponding to three different movement regimes. In Fig. 25 (a2) and (a3), the fluctuating pressure of the fixed cylinder is dominated by the lift dipole, while the drag dipole has a small intensity, which is consistent with the findings of Inoue and Hatakeyama [25]. However, the drag dipole  $A_1$  is significantly enhanced at  $\theta = 180^\circ$ , as shown in Fig. 25 (b3) and (c3). Moreover, quadrupole  $A_2$  sources also appear. The same phenomenon was also discovered by Ma et al. [40]. In Fig. 25 (b), in the direction of  $\theta = 90^\circ$ , the lift dipole  $B_1$  is a single sinusoidal curve, added to the monopole  $A_0$  and quadrupole  $A_2$ . This superposition leads to the instantaneous fluctuating pressure curves becoming narrow at the bottom and wide at the top. At  $\theta = 45^\circ$ , the noise radiated from the forced oscillating cylinder differs from the VIV. The contributions of the quadrupole and monopole are almost equal and approach that of the lift dipole. As a result, the noise radiated from a moving cylinder is obviously different from that of a fixed cylinder, and both drag dipole and quadrupole sources appear. The directivity and sound pressure of the forced oscillating cylinder and that undergoing VIV are not exactly the same, which may be due to the phase difference in the displacement and lift force which is related to vortex shedding.

Finally, the effect of the reduced velocity  $U_r$  on noise radiated from the VIV is considered. When  $U_r = 5, 6$ , and  $7$ , the cylinders are both in the lock-in regime. As before, the acoustic radiation directivity of the cylinder is represented by  $p'_{rms}$  measured at  $r' = 75L_0$  and

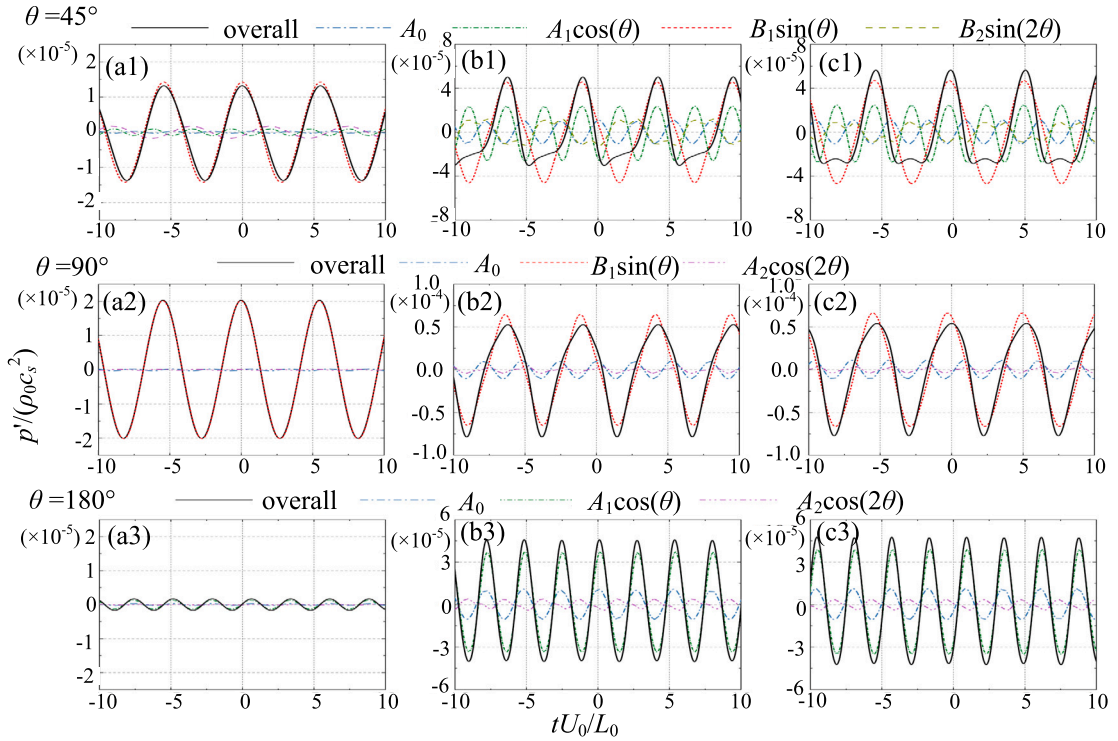


Fig. 25. A comparison of (a) fixed, (b) forced oscillating, and (c) vortex-induced vibrating cylinders in terms of their sound source contributions. The instantaneous fluctuating pressure is measured at  $r' = 75L_0$  and the decomposition is calculated at  $\theta = 45^\circ, 90^\circ$  and  $180^\circ$ , respectively.

is depicted in Fig. 26(b). It is found that the acoustic radiation directivity is almost consistent when  $U_r$  increases, and the maximum  $p'_{rms}$  are both at about  $\theta = 116^\circ$ . The noise radiated from the cylinder at  $U_r = 5$  is larger than that at  $U_r = 7$ , which corresponds to more intense vibrations. The instantaneous fluctuating pressure measured at  $r' = 75L_0, \theta = 116^\circ$  is shown in Fig. 26(a). It is found that the frequency and amplitude of the fluctuating pressure decrease as  $U_r$  increases.

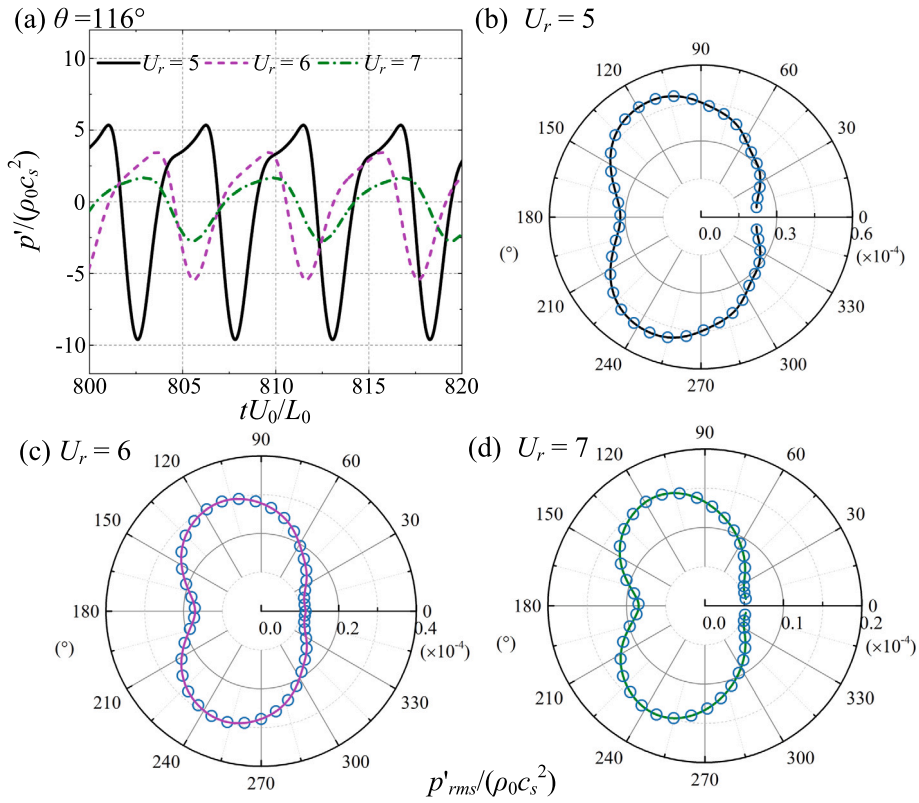
### 3.4. Computational speed tests

In this section, the performance of the proposed GICLBM in terms of the computational efficiency is demonstrated. The present GICLBM is programmed using CUDA C language and accelerated in parallel using a Graphic Processing Unit (GPU). All tests were performed on the NVIDIA GeForce RTX 3090 with a graphic memory of 24 GB. Because values of sound pressure fluctuation have a much lower level than those of flow field velocity, all variables are set with double precision. The computational domain is partitioned with structured grid cells, and the field variables are sequentially stored in arrays. An extra parameter  $T_p$  is introduced to quantify the computational speed, which is the computational time cost by each grid cell within one time step. Moreover, Million Lattice Updates Per Second (MLUPS) is also used to characterize the computational efficiency of GICLBM, and defined as [3]

$$MLUPS = \frac{(N_\theta \times N_r) \times \text{iteration steps}}{\text{running time} \times 10^6}. \quad (49)$$

As can be seen from Table 2, the calculation on an enormous number of grid points requires more graphic memory. Correspondingly, the total running time of the GICLBM increases with the graphic memory usage. The computational efficiency is basically the same on a single grid point because  $T_p$  is maintained at a fixed time. Moreover, because the VIV simulation requires an extra step to solve the equation of the cylinder's movement, the total computation time may be prolonged. However, compared with the flow field, the movement of the rigid body is solved in quite a short time, so the  $T_p$  in the three different problems are consistent and the computational efficiency is almost unaffected. For the grid used in the present work,  $512 \times 768$ , the calculation of 1000 times reference time can be completed in about two hours, indicating that the current solver is sufficiently fast to complete one case. Furthermore, the maximum amount of tested grid points is 15,728,640, corresponding to  $512 \times 30720$  and already 40 times the present grid, while the memory only occupies about 1/3 of the current device. Hence, computations on a larger domain or a finer grid are also feasible. Overall, it has been proven that the GICLBM retains the high parallel efficiency of the original LBM. It is also anticipated that the development of the GPU hardware devices will further improve the scalability and computational efficiency of the present GICLBM.





**Fig. 26.** For mass ratio  $m^* = 2$ , a comparison of the acoustic radiation directivity at different reduced velocities  $U_r$  at  $r' = 75L_0$  when  $Re = 150$  and  $Ma = 0.1$ . (a) The instantaneous fluctuating pressure curves measured at  $\theta = 116^\circ$  where the maximum  $p'_{rms}$  is located. The acoustic radiation directivity denoted by  $p'_{rms}$  with (b)  $U_r = 5$ , (c)  $U_r = 6$  and (d)  $U_r = 7$ .

**Table 2**

Comparisons of the computational speed with different grid resolution in all three scenarios.

| $(N_\theta \times N_r)$ | Memory usage<br>(GB) | Cases                       | Running time of $100T_0$<br>(hours) | $T_p$<br>(ns) | Performance<br>(MLUPS) |
|-------------------------|----------------------|-----------------------------|-------------------------------------|---------------|------------------------|
| $512 \times 768$        | 0.2                  | fixed cylinder              | 0.3                                 | 1.2           | 833.3                  |
|                         |                      | forced oscillating cylinder | 0.3                                 | 1.2           | 833.3                  |
|                         |                      | VIV cylinder                | 0.3                                 | 1.2           | 833.3                  |
| $512 \times 3840$       | 1.1                  | fixed cylinder              | 1.3                                 | 1.2           | 833.3                  |
|                         |                      | forced oscillating cylinder | 1.3                                 | 1.2           | 833.3                  |
|                         |                      | VIV cylinder                | 1.3                                 | 1.2           | 833.3                  |
| $512 \times 30720$      | 8.4                  | fixed cylinder              | 10.2                                | 1.2           | 869.6                  |
|                         |                      | forced oscillating cylinder | 10.1                                | 1.15          | 869.6                  |
|                         |                      | VIV cylinder                | 10.0                                | 1.15          | 869.6                  |

#### 4. Conclusion

In this work, a framework for calculating FIV and the associated noise radiation has been proposed. The generalized formulation of the interpolation-supplemented cascaded lattice Boltzmann method (GICLBM) on a non-uniform grid has been developed in the moving frame of reference for two-dimensional problems. Thanks to the present algorithm being based on a coordinate transformation, the far-field noise has been successfully calculated without spurious oscillation from FIV. Moreover, the perfectly matched layer (PML) technique has been embedded for absorbing the reflective noise from the boundary of the computational domain.

Under the laminar condition, the present algorithm has been sufficiently validated for computing acoustic problems using the two cases of fixed and forced oscillating cylinders. By comparing the effects of the grid resolution and non-reflective sound boundary layer thickness on the fluid dynamics and instantaneous fluctuating pressure, an appropriate grid setup has been determined. Beyond the fluid dynamics, the instantaneous fluctuating pressure and the acoustic radiation directivity have also been compared with results from other researchers, and the results have been found to be in good agreement.



The characteristics of the far-field noise radiated from a circular cylinder undergoing vortex-induced vibration (VIV) have been investigated using the efficient and accurate GICLBM, and some interesting phenomena have been discovered. Firstly, since  $Ma$  number is an important similarity criterion in computational aeroacoustic problems, the effect of  $Ma$  on the vibration is first inspected. It was found that there is a critical value at around  $Ma = 0.1$ . Secondly, the instantaneous fluctuating pressure is compared with those of fixed and forced oscillating cylinders. Monopole, drag dipole, and quadrupole sources are observed in the far-field noise. Finally, the influence of the reduced velocity ( $U_r$ ) is considered. The acoustic radiation directivity is basically similar for different  $U_r$ . However, the amplitude of the sound pressure decreases as  $U_r$  increases, and the variation trend is found to be consistent with the vibration amplitude.

Finally, the GICLBM is implemented on a GPU and its computational efficiency is demonstrated. Since the GICLBM preserves the two main steps of particle collision and streaming, it retains the high parallel performance of the original LBM. The framework proposed in the present study is based on several successful algorithms, which have both two-dimensional and three-dimensional forms [61]. Although the present algorithm is developed in the two-dimensional framework, in principle, it can be generalized into the three-dimensional formula. In the future, the framework proposed herein will also be developed for more complicated cases such as the turbulent situation, so as to provide a simple, reliable, and fast alternative for fluid-structure-sound interactions.

### CRediT authorship contribution statement

**Jian Song:** Data curation, Formal analysis, Methodology, Software, Validation, Visualization, Writing – original draft. **Haibao Hu:** Supervision. **Guanghua Li:** Project administration, Resources. **Heng Zhang:** Data curation, Formal analysis, Methodology, Validation. **Feng Ren:** Conceptualization, Data curation, Formal analysis, Funding acquisition, Methodology, Project administration, Resources, Writing – review & editing.

### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

### Data availability

Data will be made available on request.

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### Appendix A. Multi-pole decomposition of sound pressure

The fluctuating pressure is a linear superposition of noise from different sound sources. Hence, as proposed by Inoue and Hatakeyama [25], the fluctuating pressure  $p'$  can be represented by the superposition of typical sound sources,

$$\begin{aligned} p'(r', \theta, t) = & A_0(r, \theta, t) \\ & + A_1(r, \theta, t) \cos(\theta) + B_1(r, \theta, t) \sin(\theta) \\ & + A_2(r, \theta, t) \cos(2\theta) + B_2(r, \theta, t) \sin(2\theta) \\ & + \dots, \end{aligned} \quad (50)$$

where

$$\begin{aligned} A_0 &= \frac{1}{2\pi} \int_0^{2\pi} p'(r, \theta, t) d\theta \\ A_n &= \frac{1}{\pi} \int_0^{2\pi} p'(r, \theta, t) \cos(n\theta) d\theta \\ B_n &= \frac{1}{\pi} \int_0^{2\pi} p'(r, \theta, t) \sin(n\theta) d\theta. \end{aligned} \quad (51)$$

The terms on the left-hand side of Eq. (50)  $(A_0)$ ,  $(A_1, B_1)$ , and  $(A_2, B_2)$  are known as the monopole, dipole, and quadrupole. Considering the Doppler effect on sound propagation, the observer point's location  $(r, \theta)$  satisfies  $r = r' (1 + Ma \cos \theta)$  and the above equations can be modified as

$$\begin{aligned} p'(r', \theta, t) = & A_0(r', \theta, t) \\ & + A_1(r', \theta, t) \cos(\theta) + B_1(r', \theta, t) \sin(\theta) \\ & + A_2(r', \theta, t) \cos(2\theta) + B_2(r', \theta, t) \sin(2\theta) \\ & + \dots, \end{aligned} \quad (52)$$

where

$$\begin{aligned} A_0 = & \frac{1}{2\pi} \int_0^{2\pi} p'(r', \theta, t) d\theta \\ A_n = & \left(\frac{r'}{r}\right)^n \frac{1}{\pi} \int_0^{2\pi} \left(\frac{r'}{r}\right)^n p'(r', \theta, t) \cos(n\theta) d\theta \\ B_n = & \left(\frac{r'}{r}\right)^n \frac{1}{\pi} \int_0^{2\pi} \left(\frac{r'}{r}\right)^n p'(r', \theta, t) \sin(n\theta) d\theta. \end{aligned} \quad (53)$$

The sound sources  $(A_0)$ ,  $(A_1, B_1)$ , and  $(A_2, B_2)$  are calculated by integrating along  $\theta$  in Eq. (53), and then the sound sources contribution at the specified  $\theta$  can be obtained.

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